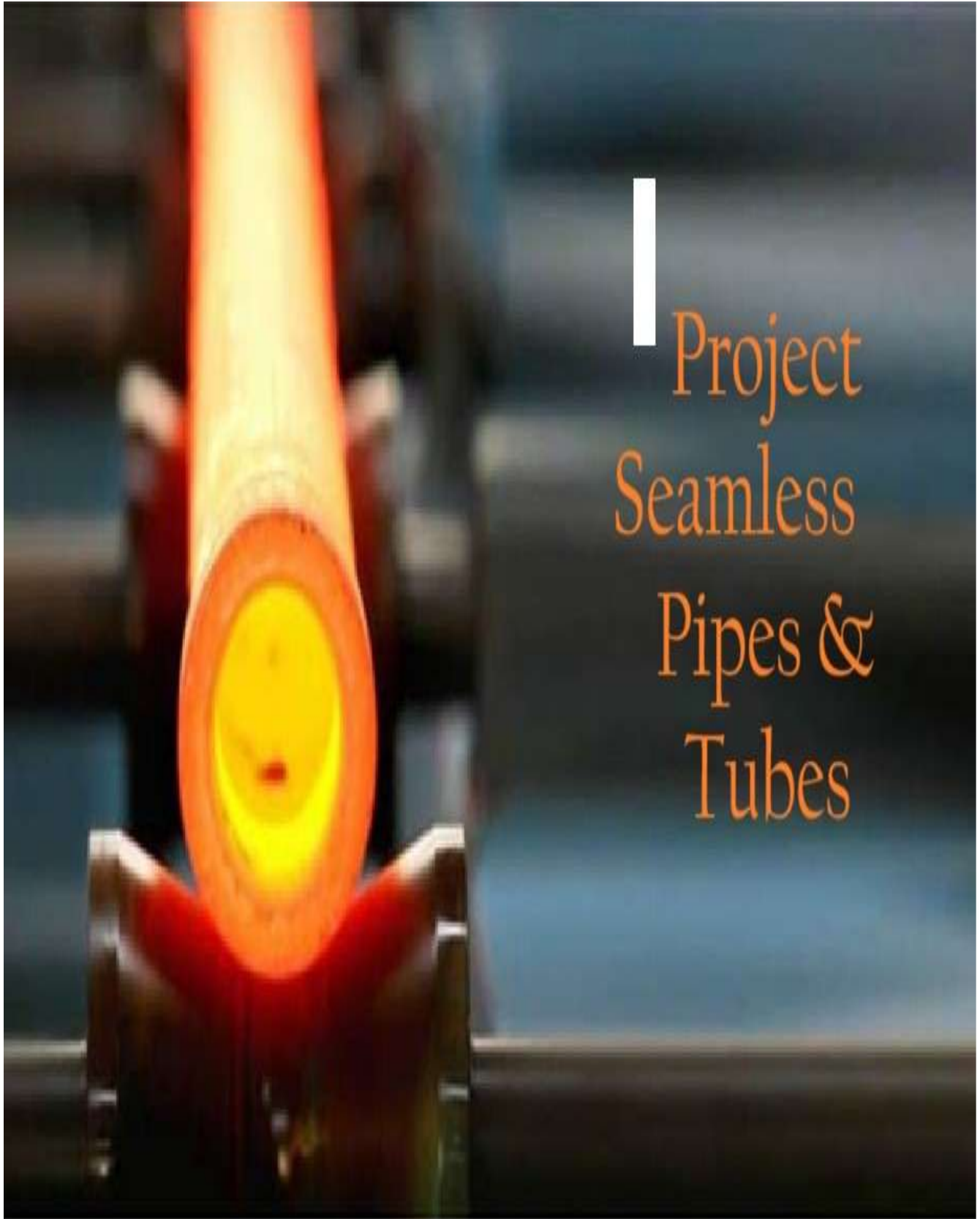




# Project Seamless Pipes & Tubes

## Introduction:

Seamless Hollow pipe



## Seamless Pipes & tubes

A Seamless tube is so named because it is made from a single piece of metal. Hence, it Seamless Pipes & tubes does not have any type of joint or 'seam' markings. They are manufactured from a metallic ingot (steel, copper, aluminium, brass, etc.) and can be hot-rolled or cold-rolled. Techniques such as punching or forming are often used with the same result: a tube without any welding.

Seamless tube is a high-tensile steel pipe used under extreme conditions such as as those at oil fields and in the transportation of gas. Made by boring a hole through a hot billet and then rolling and extruding it, seamless tube is stronger than welded pipe made by fusing together the steel plates and rolling them into the form of pipe. The seamless tube is finished to dimensional and wall thickness specifications in sizes from 1/8 inch to 26 inch outer diameter (OD). They are also used for water pipelines and boiler tubes.

The two major product segments within seamless tubes include:

- Oil country tubular goods (OCTG)
- Boiler tubes

### Oil Country Tubular Goods (OCTG):

OCTG is a family of seamless rolled products

## Oil rigs & pipes



consisting of drill pipe, casing and tubing. OCTG is largely consumed by the oil and gas exploration and production industry.

Corrosion resistance under sour service conditions is a very important OCTG characteristic, especially for casing and tubing.

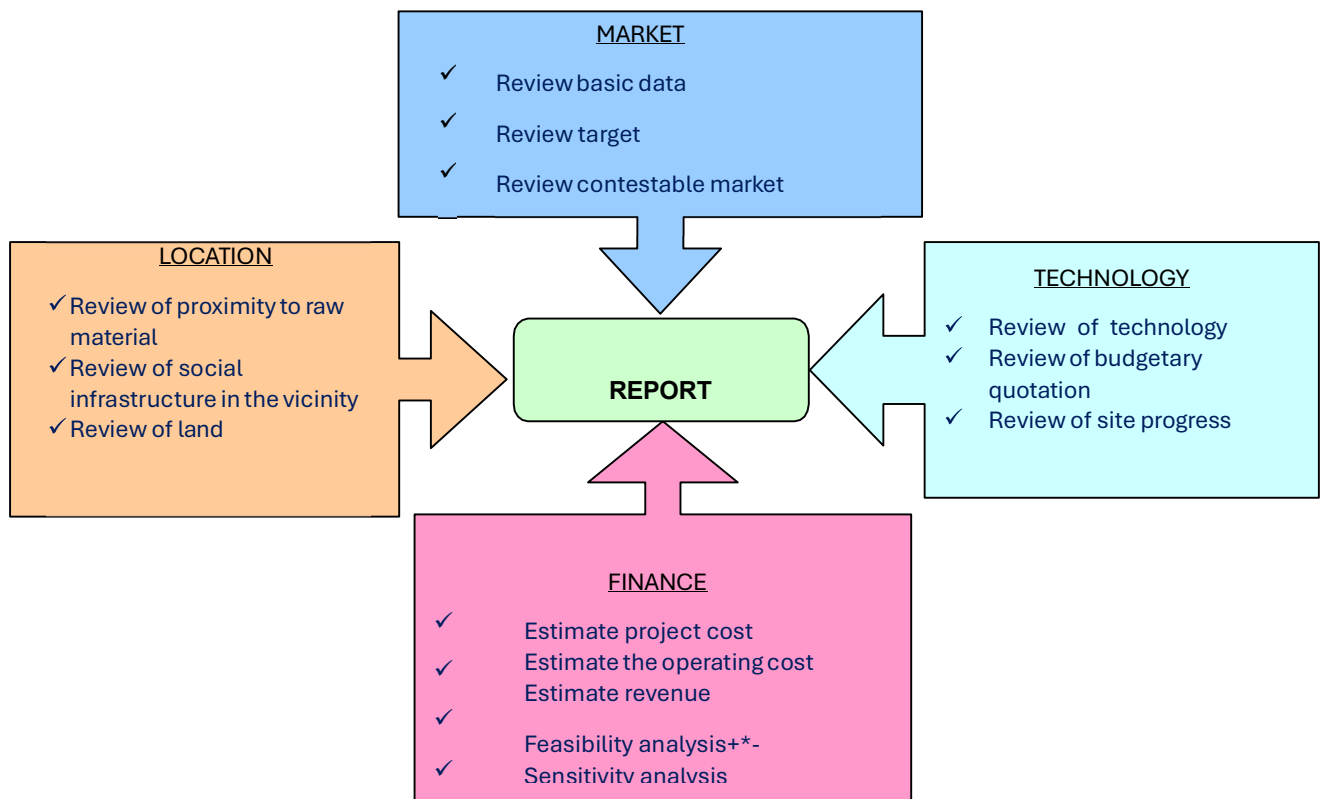
- Drill pipe is heavy seamless tube that rotates the drill bit and circulates drilling fluid. Drill pipe is simultaneously subjected to high torque by drilling, axial tension by its dead weight, and internal pressure by purging of drilling fluid. Additionally, alternating bending loads due to non-vertical or deflected drilling may be superimposed on these basic loading patterns.
- Casing pipe lines the borehole. It is subject to axial tension by its dead weight, internal pressure by fluid purging, and external pressure by surrounding rock formations. Casing is particularly exposed to axial tension and internal pressure by the pumped oil or gas emulsion.
- Tubing is pipe through which the oil or gas is transported from the wellbore

Seamless OCTG tubes/pipes are mostly used for oil and gas exploration and production as the seamless process can achieve higher grades and heavier wall thicknesses than welding can.

### **Boiler tubes**

Seamless tubes made out of alloy steel billets. These are used for high end super critical applications such as in boilers, heat exchangers, super heaters, condensers, structural and general engineering.

## Approach and Methodology:



## Seamless Tube Industry Overview:

Over the past few years, the demand for high-performance pipes has increased owing to the expansions in the energy and manufacturing sectors. This, coupled with the rising modernization of aging pipes, is expected to drive the market. Owing to these seamless pipe market trends, the market growth is expected to increase in the forecast years.

Another factor responsible for driving the market is rapid industrialization. Increasing use of high-corrosion-resistant pipes in the seamless pipe manufacturing process is further driving the market.

The demand for these pipes is likely to increase as they can be used in harsh environmental conditions, such as extremely cold or extremely hot.

Furthermore, these pipes are used in several applications, which include ship building, pipeline infrastructure, and others. Considering these factors, the market is anticipated to rise considerably during the forecast horizon.

Seamless tubes are a specialized product category with explosive demand in the rapidly expanding steel industry. Several industries, including oil & gas, construction, automotive, and manufacturing, depend heavily on seamless tubes. There is a growth in need for Seamless tubes in the plumbing, heating, and structural industries due to robust growth of the building industry in both residential and commercial projects.

In addition, seamless steel tubes are widely used in automotive applications, especially in exhaust systems where they provide greater performance, high-temperature resistance, and increased fuel efficiency.

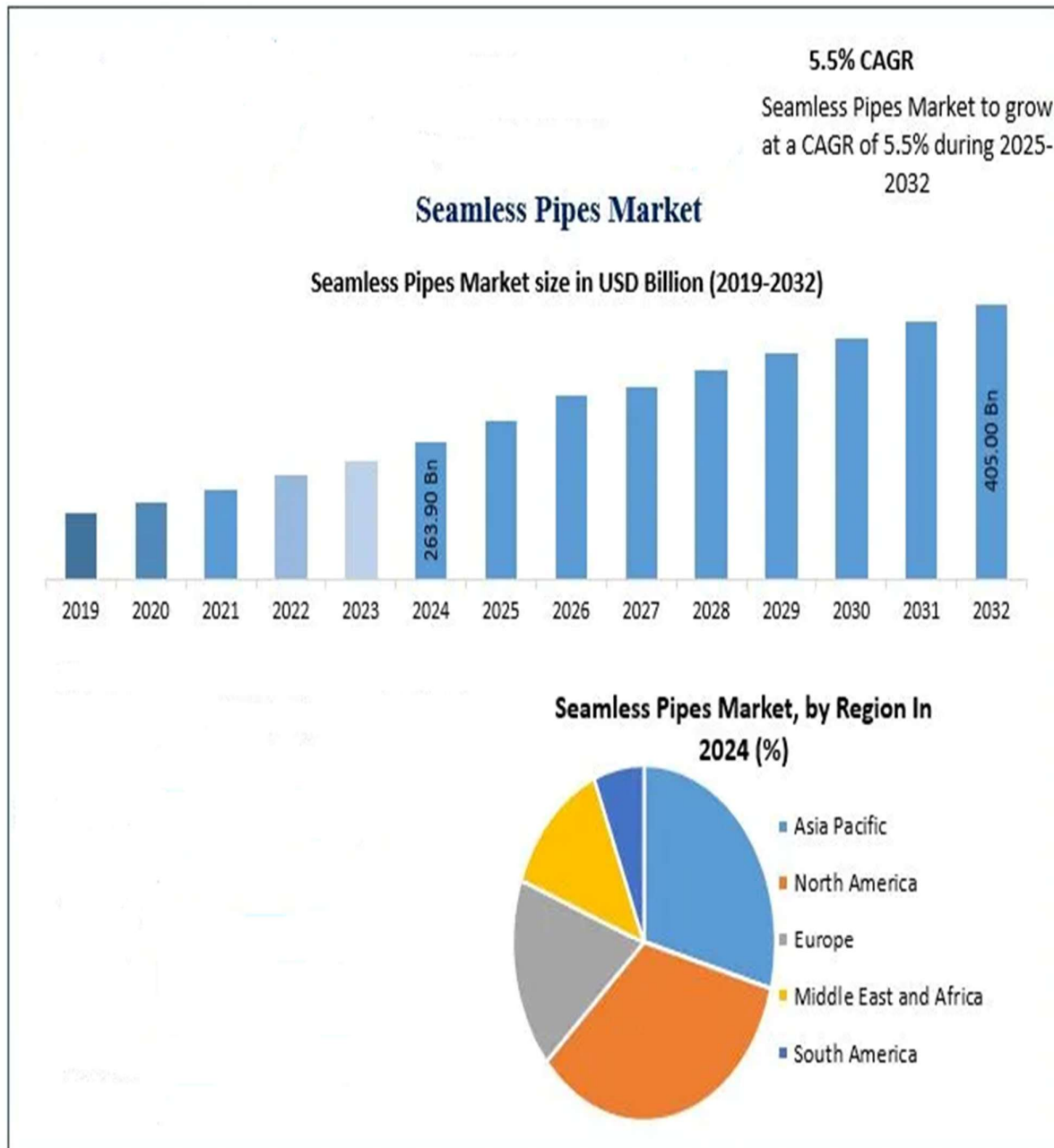
Furthermore, the need for seamless steel tubes in battery enclosures and other EV components is being fueled by the growth in focus on electric cars (EVs). Price fluctuations for raw materials such as iron ore and scrap metal have a direct effect on the total cost of manufacturing seamless steel tubes.

Steel producers have invested in innovative furnace technologies (i.e. electrical operated furnaces) that reduce greenhouse gas emissions and boost energy efficiency. In response to growth in environmental concerns and regulatory efforts to reduce emissions. Such trends are expected to offer remunerative opportunities for the expansion of the seamless steel tubes market during the forecast period.

## **CONCLUSION**

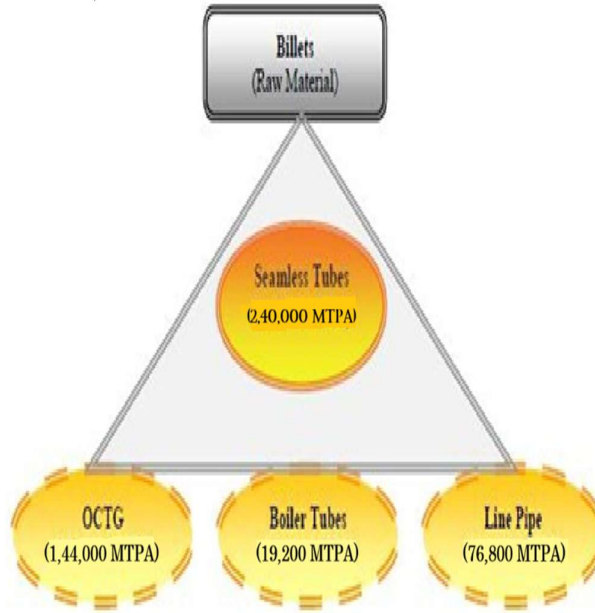
- ▶ The **World Seamless Pipes Market** size was valued at USD 263.90 Billion in 2024 and the total Seamless Pipes revenue is expected to grow at a CAGR of 5.5% from 2025 to 2032, reaching nearly USD 405.00 Billion.
- ▶ The **Middle East Seamless Steel Pipes and Tubes Market** Size is expected to experience steady growth from 2024 to 2030, largely driven by increasing demand in the energy, oil and gas, and industrial sectors. Valued at approximately USD 3.8 billion in 2024, the market is projected to grow at a CAGR of 5.8% during the forecast period, reaching an estimated value of USD 5.4 billion by 2030.





### Project Configuration

In this section the Consultants have examined the key technical features of the Project. Figure indicates the configuration of the project.



### Manufacturing Process

***Seamless tubes are corrosion-resistant, carry high tensile strength and are thus preferred in major industries for stringent applications***



Seamless tubes, unlike welded tubes, are preferred for applications where strength, resistance to corrosion, microstructure and product life is important. Ultra high strength and corrosion-resistant properties make these perfect for oil and gas industry, steam boilers, chemical and other processing industries, pipelines, installation with high and supercritical steam and pressure conditions.

Seamless tube manufacturing uses hot rolled round bar billet of carbon steel or steel alloy. The billets are cut into suitable length depending on the required length of the finished tube. The billets are heated in a furnace at a temperature of 1,200°C to 1,300°C. Then the seamless tube is formed by drawing a solid billet over a piercer to create the hollow shell (mother blanks). These hollows are crimped at one end in hot condition and then they are air cooled.



## Processes

The hot formed tubes are used for majority of the applications such as OCTG and bearings after further finishing operations such as heat treatment, threading and upsetting.

The tubes are then surface treated (pickled, phosphate coated and lubricated) to facilitate the next process of tube drawing. Depending on the finished sizes required, the tubes are subjected to single, double or triple drawing. After every draw the tubes are annealed at a temperature of 750°C. The tubes are then straightened and cleaned if necessary and before being cut to the exact size required. The finished tubes are tested for quality and they are marked and coated with rust preventive oil before being bundled and dispatched.

Several processes exist for the elongation of the hollow bloom obtained after piercing. These are listed as –

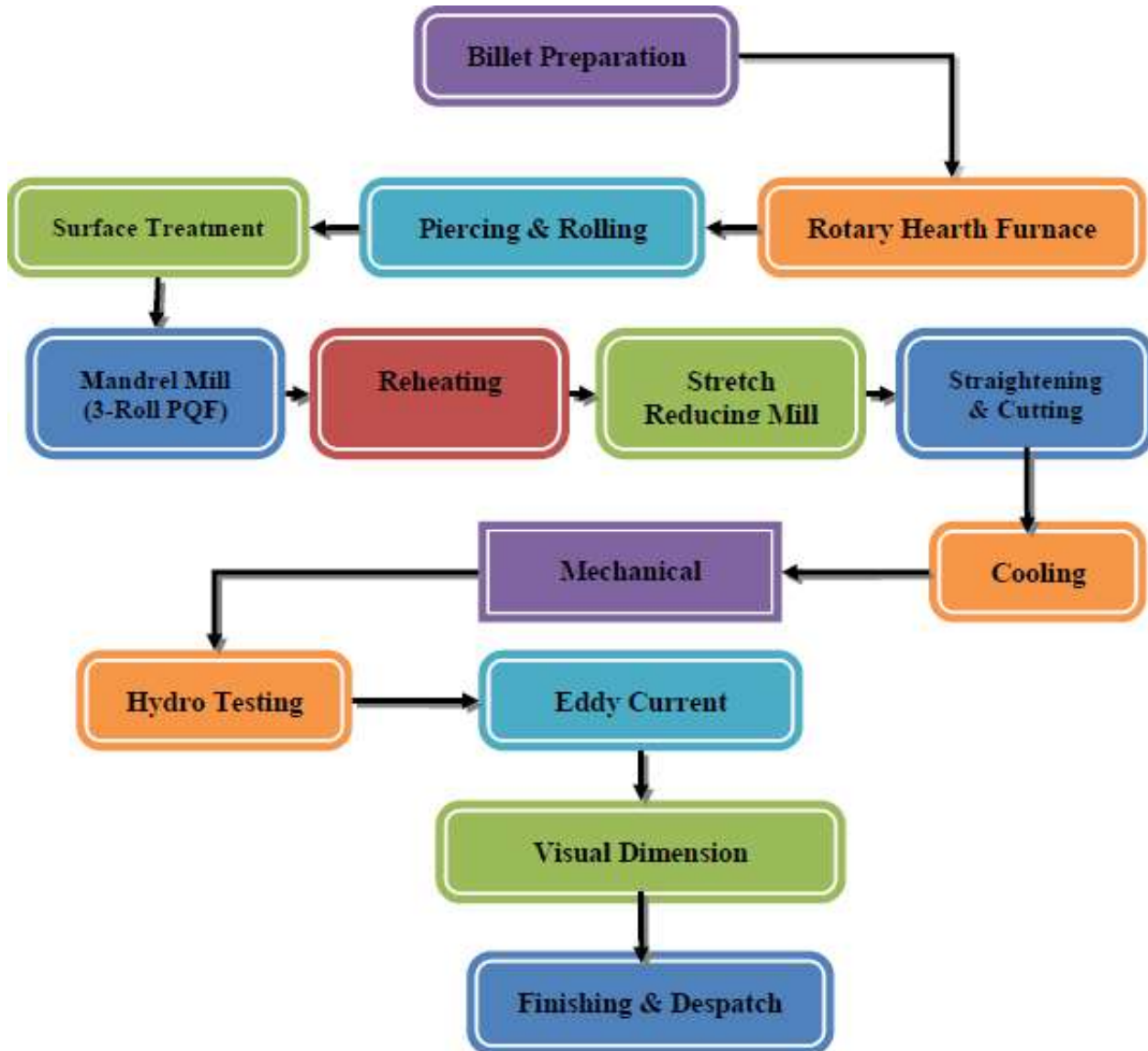
- Plug Mill
- Push Bench and CPE
- PQF (Premium Quality Finishing)
- Draw benches/Pilger Mill
- MPM (Multi-stand Pipe Mill)
- Assel Mill
- Mandrel Mill
- PSW or 3-roll Planetary Mill
- Extrusion Press

The above processes are used in the hot condition. The hot formed tubes are used for majority of the applications such as OCTG and bearings after further finishing operations such as heat treatment, threading and upsetting. For certain applications, tubes of smaller diameter, higher mechanical properties or better surface finish are required. Such tubes are obtained by cold processing of the above hot formed or mother tubes.

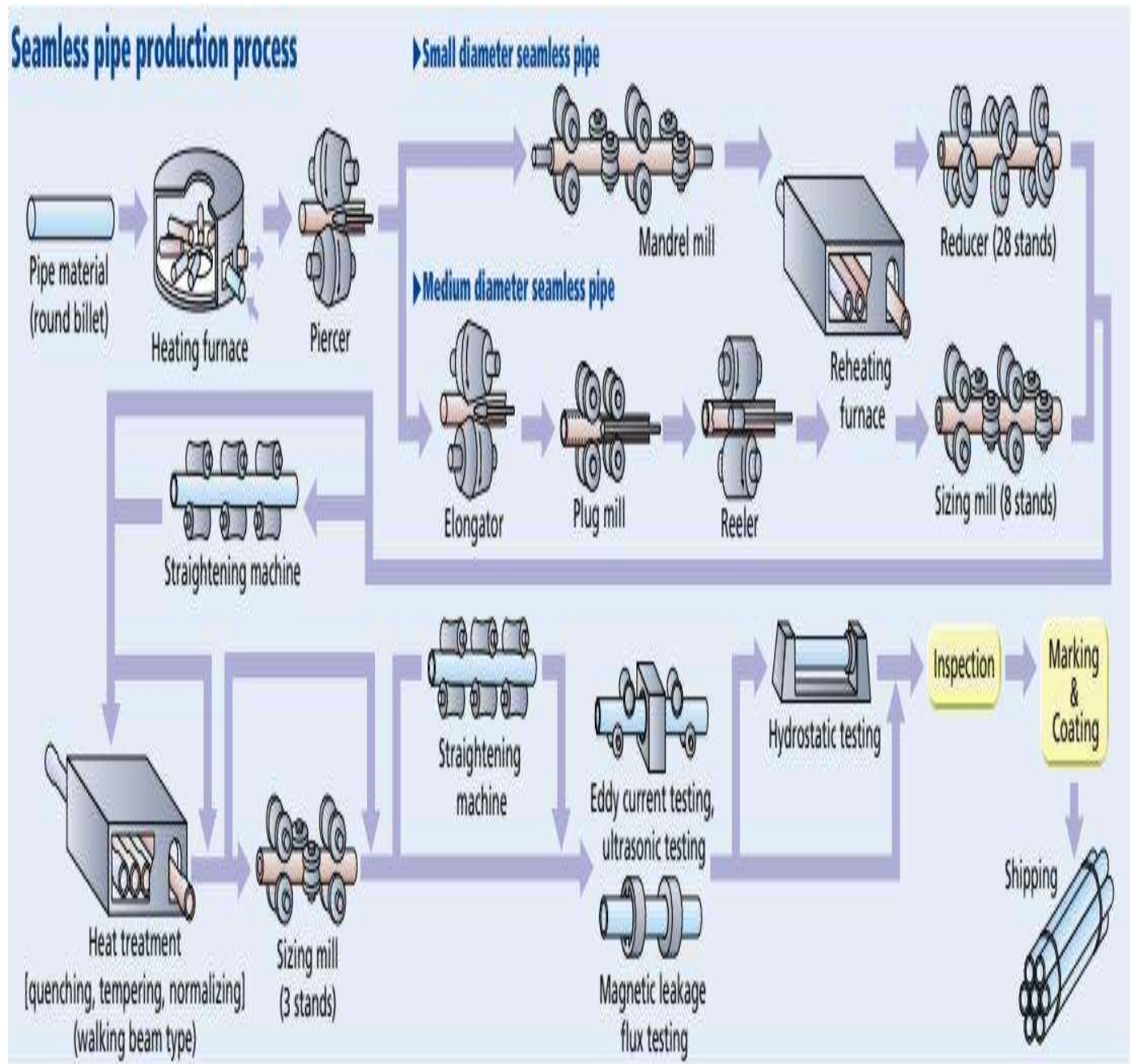
There also exist processes on the horizon such as Hydrostatic Extrusion, Continuous Casting and Rolling and Powder Metallurgy which could be used in special cases for manufacturing of seamless tubes. However, for commercial manufacturing of seamless tubes in bulk quantities, these processes do not have much relevance at present.

### Process Flow

The process flow diagram for the production of seamless tubes is shown in figure below.



## Process Flow



## Process

Rotary hearth furnaces are commonly used to heat steel billets in the production of seamless pipes. These furnaces offer advantages like high capacity, good temperature uniformity, and ease of automation

### Billet Preparation

Round billets are used as input raw materials. These pipe billets are processed into qualified gauges ,billets through the operating sequences such as transfer, saw cutting inspection and repair grinding in the billet preparation area and then transferred into the rotary hearth furnace for heating .

Rotary Hearth Furnace



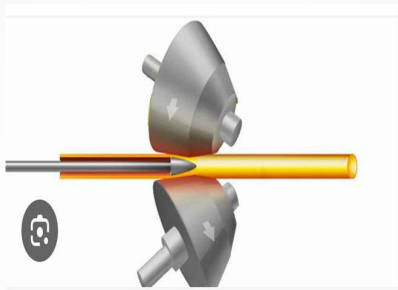
### Piercing and Rolling

The billets, heated to a temperature range of 1,200°C to 1,300°C, are pierced over the piercer with combined rotational and axial advance motion. The stress generated induces an opening tendency at the billet center. By positioning a suitable mandrel, advantage of this tendency is taken to form a hollow core with controlled internal and external diameters.



## Process

Seamless pipes are produced using piercing rolling process in which round bars are fed between two rolls and pierced by stationary plug



The design of the roll pass and roll speed control of the mandrel and roll of the three-roll continuous pipe mill with restrained mandrel force, allows the rolled piece to flow longitudinally, reducing the wall thickness to the desired dimensions. The last two stands makes pipe and have small deformation, with their main function of ensuring dimensions of pipe.

### Surface Treatment

Before rolling on the three-roll continuous pipe mill with mandrel retained system, the hollow shell is crimped at one end in hot condition and then air cooled. Anti-oxidizing powders are blown into the inner holes of the pierced billets for de-scaling internally by adopting nitrogen. Tubes are then pickled, phosphate coated and lubricated to facilitate the next process of tube drawing.

During the run of the mandrel return roll table the mandrel is cooled through three groups of tunnel-type coolers for first time cooling.

### Premium Quality Finishing (PQF)

After the piercing process is complete the hot hollow bloom is rolled

## Process

PQF technology offers improved production efficiency, yield, quality finish, strength and reduced redundancy to the seamless tube manufacturing process

on the three-roll continuous pipe mill with mandrel retained system. This process is termed as 3-roll PQF.



In this process the hollow bloom undergoes elongation over three stages. This ensures stage wise dimension control reducing the redundancy and ensuring higher tolerance of the tube. This process shortens the rolling rhythm and increases the productivity of manufacturing.



## Process

Reheating seamless pipes before sizing is crucial for several reasons, primarily to improve the material's plasticity, control its microstructure, and ensure desired mechanical properties

### Reheating

After the PQF process, tubes are reheated at 750°C. This is done so as to ensure stress reduction and induce effective tolerance in the tubes.



### Stretch Reducing Mill

After elongation, hollow tubes are forwarded to the stretch reducing mill. Here tubes are given the final touch for bringing them out in the

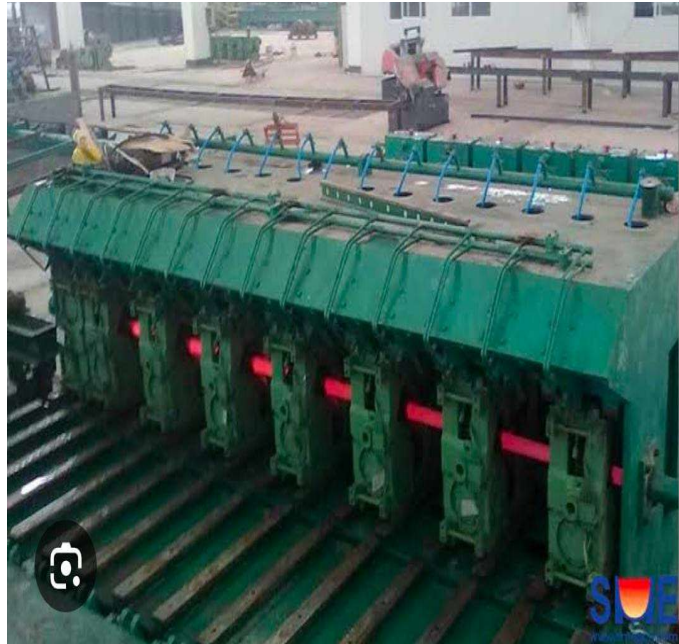
## Process

Seamless pipes are sized using a sizing mill stand (SRM) in the final stage of their production, following hot rolling



desired shape and dimensions. This stage also ensures diminishing the possibilities of stretch in the obtained tube.

### Sizing Mill



### Straightening & Cutting

Tubes are then straightened and cleaned (if necessary) before cutting to exact required size.

## Process



### Cooling

In order to enhance tensile strength and improve quality the tube undergoes cooling. Here the hot bloom is air and water cooled which drastically reduces the temperature of the tube thereby preparing it for quality testing.

### Quality Testing

These tubes then undergo various quality tests viz. hydro test and eddy current test so as to ensure the quality of the tube.

These tubes are then threaded, coupled screwed followed by a drift test. These are then marked and coated with rust preventive oil and sent for bundling and despatching.

### Process Technology

When the hot bloom pierced billet is obtained after piercing, it is further elongated into a shell. This is an important step in the entire process for manufacturing seamless tube. Some of the well-known process technologies for this critical stage are listed as –

- Plug Mill
  - Push Bench and Cross Roll Piercing & Elongation (CPE)
  - PQF (Premium Quality Finishing)
  - Diescher Mill
  - Pilger Mill
  - MPM (Multi-stand Pipe Mill)
  - Assel Mill
  - Mandrel Mill
  - PSW or 3-roll Planetary Mill
  - Extrusion Press
- Plug Mill is preferred for manufacturing higher diameter seamless tubes.
- Push bench and CPE technologies are used for commercial tubes and for cold drawing blanks. They are used for the production of smaller outer diameter seamless tubes like those of 7” and nearby.
- PQF is said to be the 21<sup>st</sup> century breakthrough in the production of seamless tubes. This technology enhances the production and the quality output.
- The Diescher process which had become obsolete has been revived in recent years as a flexible, low investment process for

## PQF

PQF is becoming one of the most popular process technologies being adopted by the industries for the production of seamless tubes

manufacture of mechanical tubing of smaller diameters

- The Pilger Mill is suitable to roll tubes in the size range of 150-600 mm. It is a flexible process with low initial investment.
- The Multi-stand Pipe Mill (MPM) has been developed for high production of medium diameter tubes at minimum cost. The process takes advantage of high scale production to reduce unit cost of production.
- The Floating Mandrel Mill is used for production of smaller diameter tubes.
- The Assel Mill has been particularly popular for manufacture of bearing and mechanical tubes in medium capacity range. A number of developments have taken place in Assel Mill to improve process efficiency.

– The Planetary Mill has been adopted for manufacture of tubes.

Although a number of advantages are claimed, the process technology is complex and has not been popular.

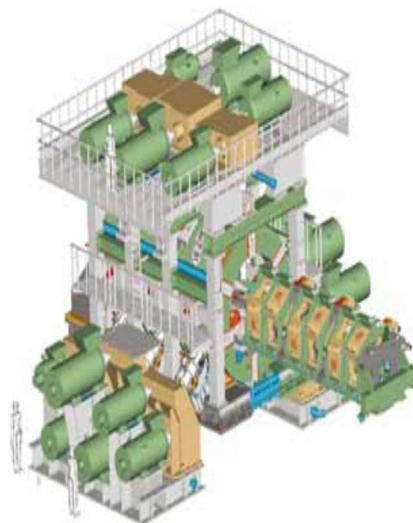
– Manufacture of seamless tubes by extrusion is well-established

process technology. It has certain advantages with respect to quality. However, the production costs are higher and hence extrusion is used generally for manufacture of stainless steel and high alloy steel tubes or nickel base alloy tubes, which fetch higher prices.

PQF is becoming one of the most popular process technologies being adopted by the industries for the production of seamless tubes.

### Premium Quality Finish (PQF) Technology

The PQF has is a longitudinal rolling mill, consisting of a number of adjustable, driven, 3-roll type stands, which uses the retained mandrel technology for the rolling of the mother tube. The deformation which takes place in the 3-roll pass is highly uniform leading to a great improvement in the tube quality, a consistent reduction of production cost and a high productive flexibility.

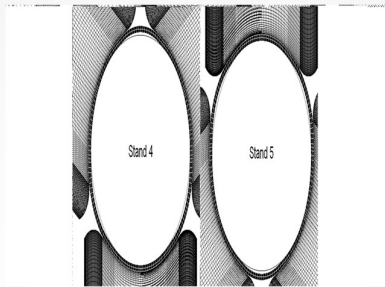
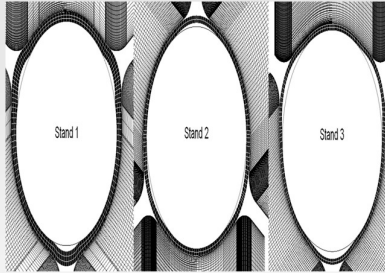


The rolling unit consists of three driven rolls assembled in a



cartridge. Each roll is supported by chocks fitted to a lever type holder. The lever is pivoting on a pin, fitted in the cartridge. For the roll changing, the assembled system is rotated outside the cartridge where the chocks are disconnected from the levers. Therefore, the levers always remain assembled on the pin in the cartridge.

### PQF



### Advantages of PQF Technology

PQF technology is slowly becoming popular in the seamless tubes industry. The main features which differentiate it from other technologies are –

- Improved production efficiency and yield
- Better quality of finished tubes
- Higher tensile strength of the tubes
- It reduces the redundancy of dimensions ensuring higher improved finish as compared to other processes
  - It lowers the longitudinal tension and material stretching
  - It has a lower difference of the peripheral speed between the groove bottom and the flange of the rolls resulting in a far more uniform deformation in the pass, both in steady and in transient conditions
- It reduces the amount of residual deformation which is conducive to rolling high grade alloy steel tubes
- It has a reduced flange section (about 30% less compared to the 2-roll), which allows a better control of the metal flow towards the flange. The benefit becomes particularly important during the tail end rolling, when the metal flow is not restrained by the material at the stand entry as in the steady state condition. Therefore, the end collapse and the formation of fins is avoided or drastically reduced
- In this process the average unit rolling pressure required is low ensuring a uniform metal deformation thereby lowering the temperatures rolling. This controlled rolling makes the normalization, quenching and other processes become possible online
  - It has a compact overall design with smaller foundation depth
  - The space required to assemble this unit is small
  - It enables quick size change over

Possible hurdles of PQF technology adaptation in the backdrop of its selection by the Client can be highlighted as –

- For the fact that the technology is new trial run might exceed the proposed time till the success for commercial run is achieved.
- As PQF is a very modern technology providing technical training to the staff might incur higher cost and time to the Client.
- As the PQF technology suppliers are limited the Client may have lower control over its supplier regarding training costs and other technology related assistance.



### **Technology Suppliers**

From discussions and secondary sources, the Consultants note the following –

- PQF technology providers are limited globally

The biggest and the first PQF technology provider globally is the German giant SMS-Meer which was earlier known as Mannesmann- Meer. Taiyuan Heavy Industry Co. Ltd., China, is another upcoming name in the technology provider of PQF.

#### **SMS-Meer, Germany**

SMS Meer is a division of the SMS group, world market leader in plant and machine engineering for the industrial processing of steel, aluminum and non-ferrous (NF) metals. SMS-Meer was earlier known as Mannesmann-Meer and is presently the biggest supplier of technology supporting seamless tube production. Technologies such as PQF, CPE, Plug mill, Pilger mill, Assel mill etc. are supplied by SMS- Meer globally. Maharashtra Seamless and ISMT – the two major producers in India – use the technology provided by SMS-Meer. SMS- Meer's global presence is backed by its operating units in numerous countries.

The greatest strengths of SMS-Meer lie in their highly qualified and experienced personnel engaged in constant technological development and the company is certified on quality standards by DIN ISO 9001.

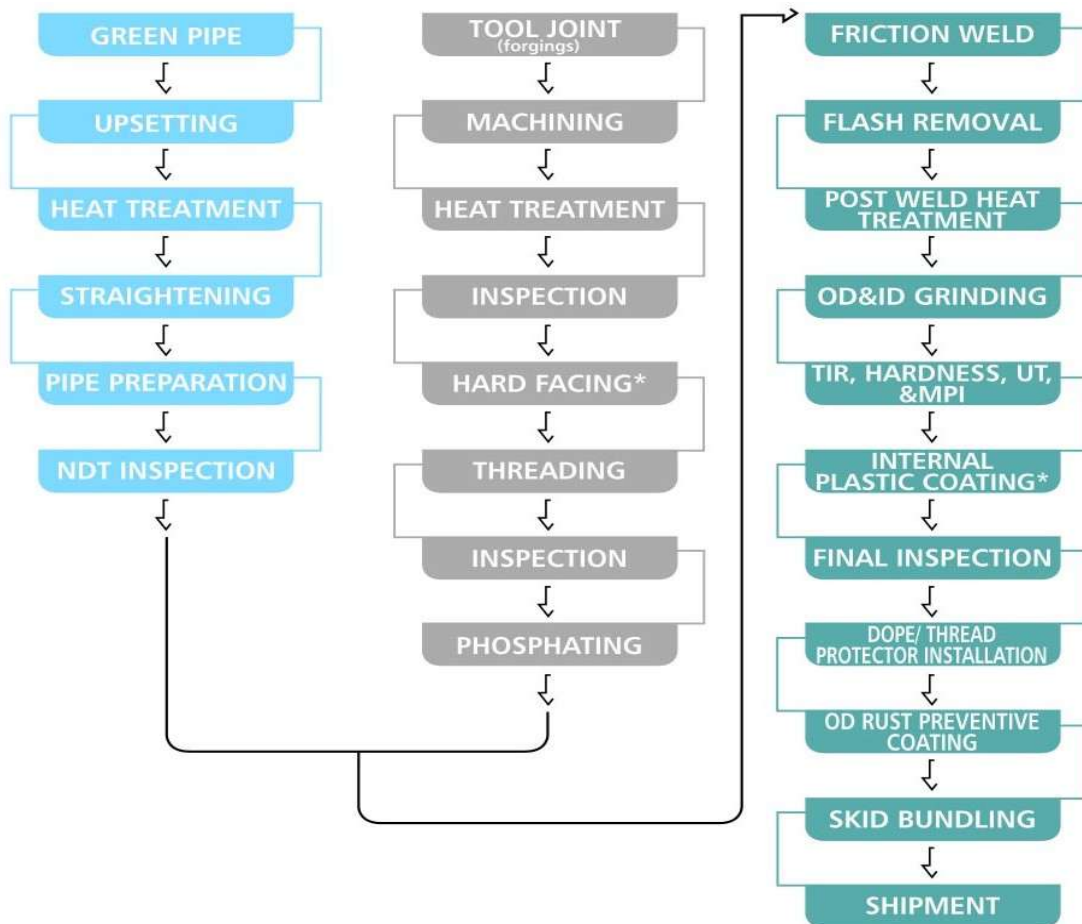
#### **Taiyuan Heavy Industry Co. Ltd.**

Taiyuan Heavy Industry Co. Ltd. (TYHI) was established in 1950. It is the first heavy machinery manufacturing enterprise designed and built since the inception of new China. TYHI is well-known both at home and abroad for becoming the first listed company in China's machine – building industry in 1998. It is one of the most renowned heavy machine manufacturing traders, the largest crane manufacturing base, the largest oil film bearing manufacturing base or heavy-duty rolling mills, the largest plate leveler (tube straightener) manufacturing base, the largest multifunctional rotating stage manufacturing base, unique train wheel and axle pair manufacturing base and the forging equipment manufacturing base with the most complete varieties in China.

The product designed and produced by TYHI have been widely used in the sectors such as metallurgy, mine, hydropower, thermal power, nuclear power, energy, aviation, space flight, chemistry, railway, ship building, environmental protection, communication and transportation. A lot of items have filled the blanks of the state. Many products have won gold medals for state quality products. It has created 340 China No.1 and world No.1 as well, known as "state's treasure" and "path breaker of national economy". Machine products have been exported to more than 20 counties and regions in the world.

TYHI has gained customers' approval and trustfulness with first-class products, first-class quality and first-class service in the process of pursuing brilliance and creating world class technologies.

## Drill Pipe Manufacturing Flow chart



### Mechanical Properties of API Drill Pipe Grades

| Group | Grade | Yield Strength |     |         |      | Tensile Strength<br>Minimum |      | Elongation %<br>Minimum | API<br>Specification<br>on |
|-------|-------|----------------|-----|---------|------|-----------------------------|------|-------------------------|----------------------------|
|       |       | Minimum        |     | Maximum |      |                             |      |                         |                            |
|       |       | PSI            | MPa | PSI     | MPa  | PSI                         | MPa  |                         |                            |
| 1     | E-75  | 75,000         | 517 | 105,000 | 724  | 100,000                     | 689  | *                       | 5DP                        |
| 3     | X-95  | 95,000         | 655 | 125,000 | 862  | 105,000                     | 724  | *                       | 5DP                        |
|       | G-105 | 105,000        | 724 | 135,000 | 931  | 115,000                     | 793  | *                       | 5DP                        |
|       | S-135 | 135,000        | 931 | 165,000 | 1138 | 145,000                     | 1000 | *                       | 5DP                        |

### Mechanical Properties of Tool Joint

| Minimum Yield Strength |     | Maximum Yield Strength |      | Maximum |     | Elongation % Minimum | API Specification on |
|------------------------|-----|------------------------|------|---------|-----|----------------------|----------------------|
| PSI                    | MPa | PSI                    | MPa  | PSI     | MPa |                      |                      |
| 120,000                | 827 | 165,000                | 1138 | 140,000 | 965 | 13                   | 285.331              |

## Drill pipe & tool joints



Plain end pipe is upsetted and heat-treated full length by the Quench and Temper process to achieve uniform mechanical properties. The upset profile is designed to blend with the inside diameter of the pipe body with a generous radius giving a smooth transition from the thicker cross section of the upset to the pipe body. The smooth transition minimizes the stress concentrations.

The heat-treated pipe is tested for tensile and impact properties and are 100% non-destructively inspected for possible longitudinal and transverse defects, wall thickness and grade verification. Dry magnetic particle inspection of the upset ends is done to ensure that the upsets are free from defects. The OD & ID and face of the upsets are CNC machined for a clean surface as part of pipe preparation prior to welding.

Tool Joint Forgings are turned, bored and faced prior to heat treatment by the Quench and Temper process to produce a uniform microstructure, high ductility, and other mechanical properties in accordance with API Specifications. The heat treatment is done under controlled atmosphere for consistent quality. The hardness of every tool joint is checked by Brinell Hardness Tester and recorded. The tensile and impact properties are verified by destructive testing of a tool joint representing each process batch. Every tool joint is subjected to wet magnetic particle inspection, both transverse and longitudinal, to insure the absence of defects.

Threads are precision machined on CNC lathes. All connections are 100% inspected for thread parameters and phosphatized either with Zinc or Manganese for antigalling treatment.

Tool Joints are attached to the drill pipe tube by friction welding to produce a high integrity solid state weld connection between the tool joint and the upset drill pipe tube. The process involves the rotation of one surface against a second surface at a relatively high speed and under heavy pressure. The friction between the tool joint surface and the pipe surface generates heat, causing the contact surfaces to reach a high temperature, below the melting temperature, at which they are forged together, producing the weld. Every weld is monitored and recorded in a strip chart.

The OD and ID weld rams are removed by shearing and machining. Post weld heat treatment of the heat-affected zone between the drill pipe upset and the tool joint is carried out to restore the properties of the weld zone. The post weld heat treatment is a Quench and Temper process by which the weld zone is first brought to its austenitizing temperature, and then polymer quenched. An area wider than the weld zone is then reheated by induction to tempering temperature. Every post weld heat treatment is monitored and recorded in a strip chart.

## Drill pipes & Tool joints



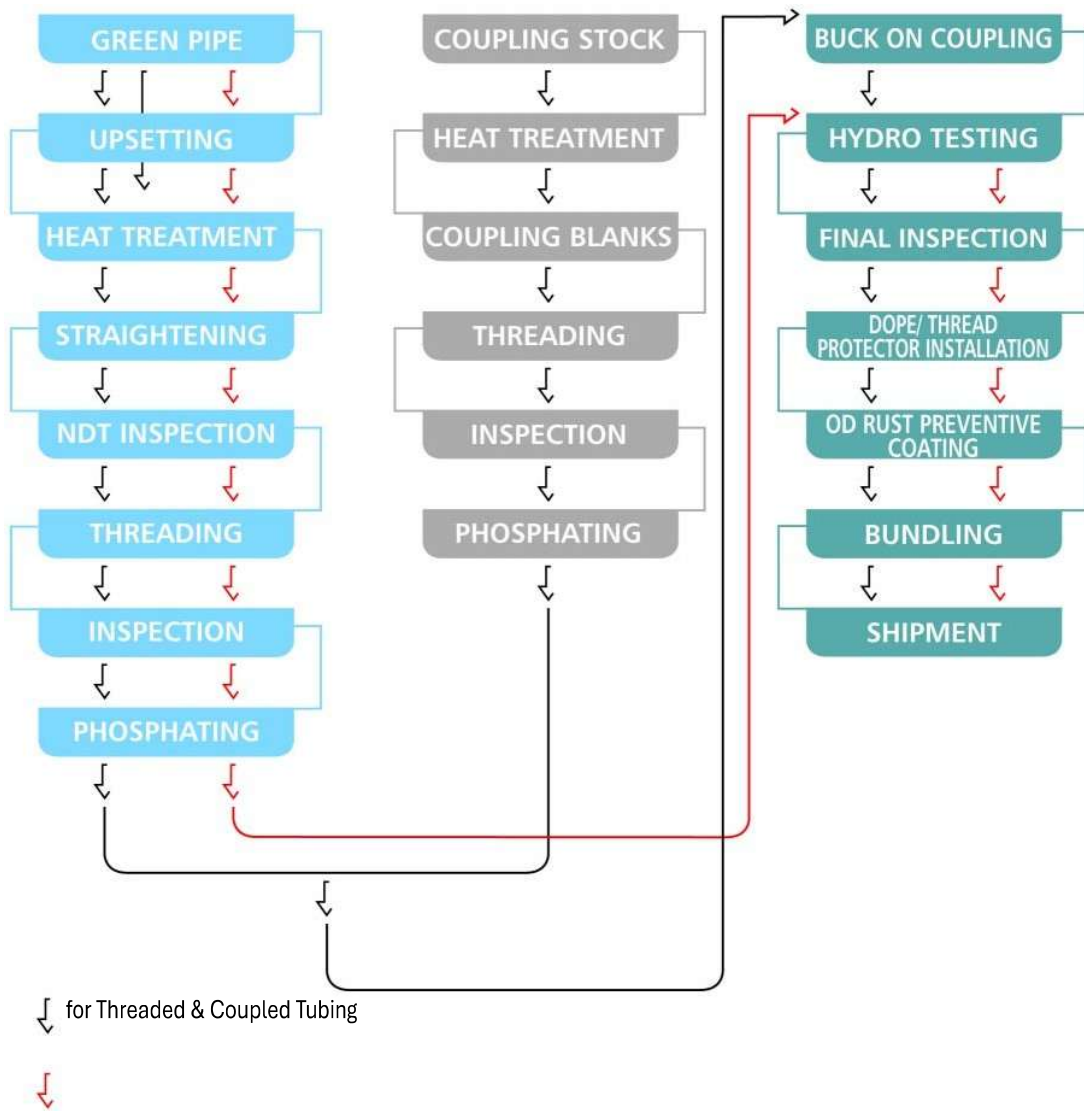
Each weld is 100% inspected for hardness to confirm that they were adequately tempered. The post weld heat treatment process consistently produces stronger, tougher, and more uniform weld properties and a desirable microstructure, which maximizes the resistance to stress and fatigue.

Surfaces adjacent to the weld line are smooth finished on both the OD and ID. Precision machining and grinding provides a surface finish that is free of stress risers. Each weld area is 100% inspected using wet magnetic particle and ultrasonic inspection techniques. Destructive testing of the weld covering tensile, yield, impact and bend tests are carried out for every defined lot and are documented.

The Quality Control involves sampling at defined and random intervals for process control.

All the tests and inspection results and product traceability data are fully documented and maintained in a computerized data base. At any time, the Customer can be provided with a comprehensive account of his product.

## Tubing Manufacturing Flow chart





### Tubing pipes



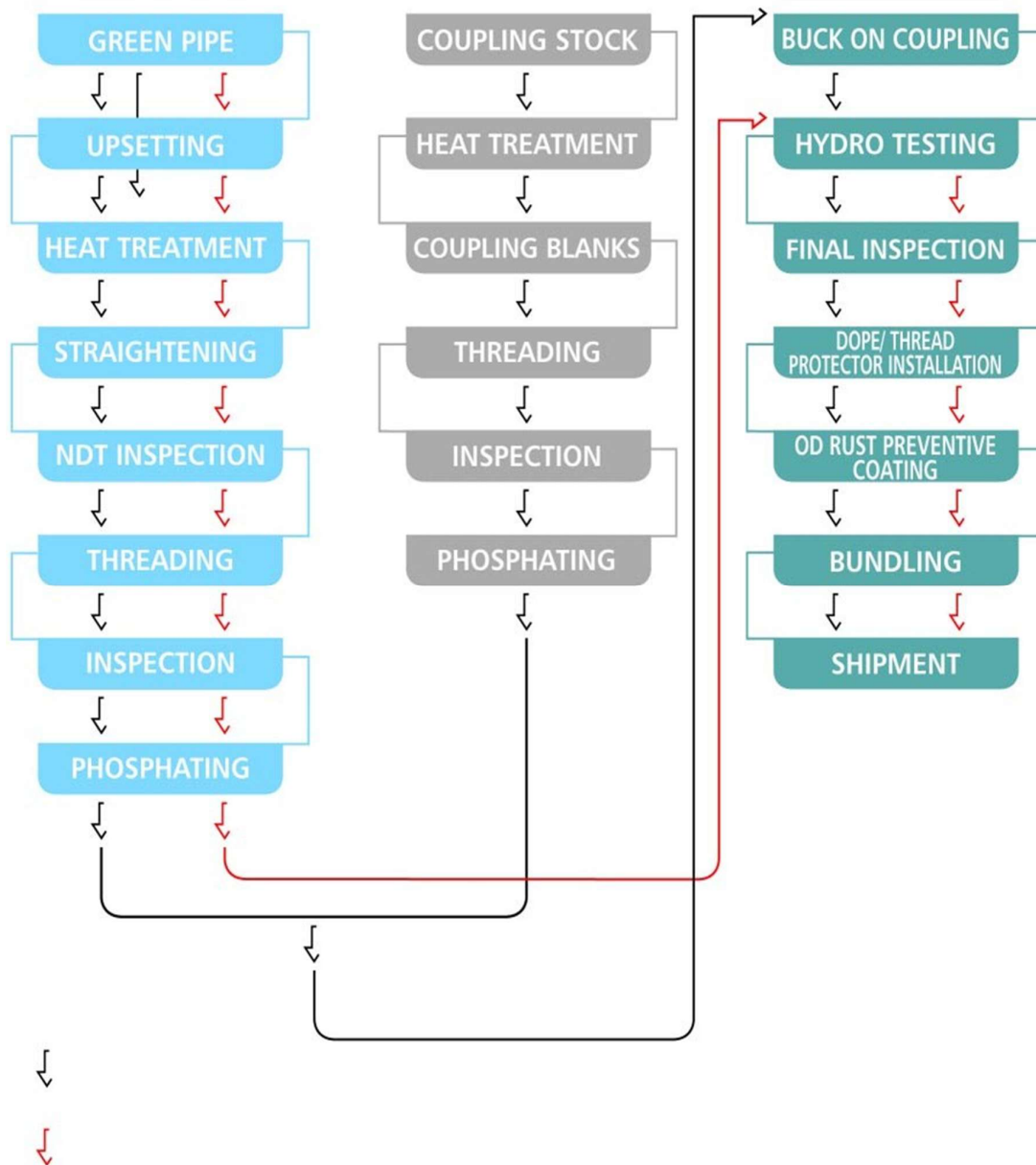
Plain End Pipe is upsetted (where required) and heat-treated by Quench and Temper process. Each Heat Lot is tested for mechanical properties. The heat treated Tubing is 100% inspected non-destructively prior to threading. The Coupling Stock follows the same process prior to threading.

The threading on the Pipe and the Couplings is done on CNC machines and 100% gaging of all the critical dimensional parameters is done and recorded as part of the manufacturing records. The finished Couplings are given antigalling treatment (Zinc or Manganese) prior to Buck-on operation. There is an on-line antigalling treatment facility with tilt tables and phosphating tanks for both ends of Premium Tubing. Computerized Torque Monitoring System is utilized for Buck-on operation.

The Couplings are Bucked-on to the pipe at their specific designed buck-on torques which are recorded. All Tubing are subjected to Hydrostatic Testing as required by API Specifications.

Internal Plastic Coating of Tubing can also be provided at customers' option. The finished Tubing is externally coated with rust preventive varnish, installed with Thread Protectors and Skid Bundled with utmost care. Complete documentation is provided with each shipment detailing the full Quality Control records.

## Casing Manufacturing Flow chart



## Casing pipes



Plain End Pipe is upsetted (for Extreme Line Casing only) and heat-treated by the Quench and Temper process. Each Heat Lot is tested for mechanical properties. The heat treated Casing is 100% inspected non-destructively prior to threading. The Coupling Stock follows the same process prior to threading.

The threading on the Pipe and the Couplings is done on CNC machines and 100% gaging of all the critical dimensional parameters is done and recorded as part of the manufacturing records. The finished Couplings are given antigalling treatment (Zinc or Manganese) prior to Buck-on operation.

The Couplings are Bucked-on to the pipe at their specific designed buck-on torques which are recorded.



All Casing are subjected to Hydrostatic Testing as required by API Specifications. The finished Casing is externally coated with rust preventive varnish, attached with Thread Protectors prior to shipment

## Environment Management

### Environment management plan

In this section the Consultants have listed environmental statutory approvals and clearances required prior to operation of the Project. The Consultants have also briefed on the likely environmental impacts, mitigation measures, guidelines and standards set by CPCB/GPCB.

### Environmental Clearances and Approvals

List of environmental statutory clearance and approvals required prior to operation of the Project and their current status are as indicated in Table

| S. No. | Environmental clearance/approvals                                                                |
|--------|--------------------------------------------------------------------------------------------------|
| 1      | Consent to Establish/NOC from the Pollution Control Board                                        |
| 2      | Approval for usage of water                                                                      |
| 3      | Approval for power usage                                                                         |
| 4      | NOC from Department of Explosives for storage of fuel for D.G set                                |
| 5      | No objection or site clearance from local authority                                              |
| 6      | Licence under Factory Act                                                                        |
| 7      | Consent to Operate under Air Act and Water Act from PCB required prior to operation of the plant |

### Environmental Impacts and Mitigation Measures

Environmental impacts caused by the project can be broadly categorised as —

- Air pollution
- Noise Pollution
- Water pollution
- Solid waste

#### • Air Pollution

The plant can cause air pollution due to the following operation:

Natural gas is the cleanest of all fossil fuels. The combustion of natural gas releases a very small amount of sulphur dioxide and nitrogen oxides, virtually no ash or particulate matter and lower level of carbon dioxide, carbon monoxide and other reactive hydrocarbons.

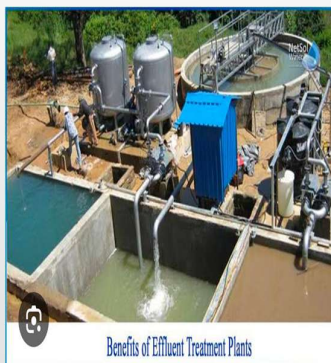
- Emission of fumes and gases from reheating furnace.
- Emission from D.G sets

▪

The Consultants suggest the following air pollution mitigation measures:

- As per the environmental standards by CPCB, NO<sub>x</sub> emission shall be restricted to 100ppm for units burning natural gas.

## Environmental impacts



- Suitably designed chimney shall be provided to rotary and reheating furnace to facilitate proper dispersion of pollutants.
- Chimney height shall be calculated using the formulae  $H = 14 Q^{0.3}$  where Q is the SO<sub>2</sub> emission in Kg/hr.
- D.G set shall be provided with adequate stack height for proper dispersion of pollutants. As per CPCB guidelines the stack height shall be calculated using formulae  $H = h + 0.2 * KVA$  where
  - H = Total height of stack in meters
  - h = Height of building where D.G set is installed
  - KVA = Total generator capacity of Set in KVA
- Thick green belt @ 33% of the total plant area shall be provided to curtail dust emission.

However the Company shall conduct continuous stack monitoring and regular ambient air quality monitoring to ensure that they comply with emission and ambient air quality standard as indicated in Table 8.2.

### Ambient Air Quality Standard

| S.No | Parameters                 | µg/m <sup>3</sup> |
|------|----------------------------|-------------------|
| 1    | SO <sub>2</sub>            | 80                |
| 2    | NO <sub>2</sub>            | 80                |
| 3    | Particulate Matter (PM10)  | 100               |
| 4    | Particulate Matter (PM2.5) | 60                |
| 5    | Carbon monoxide            | 4,000             |
| 6    | Ammonia                    | 400               |

### Water Pollution

The plant can cause water pollution due to the following operation -

- Impact due to usage of ground water
- Runoff from the raw material storage yard
- Process and utility washings
- Sewage generated due to domestic activities.

The Consultants suggest the following water pollution control measures:

- Garland drains with settling pit shall be provided at raw material stock pile area to prevent washout during rainy season.
- Process or utility wash water shall be adjusted for pH, oil and grease and sediments. Treated effluent shall be reused within the system so as to achieve zero discharge.
- Domestic effluent shall be provided with septic tank followed by soak pit or STP as per the specific condition of consent from GPCB.

Any discharge from the plant shall comply with the general effluent discharge standards as indicated in Table .



## Project- Seamless Pipes & Tubes (Dia 60.3 – 278 mm size )

| S.No | Parameter                                                    | Concentration in mg/l |
|------|--------------------------------------------------------------|-----------------------|
| 1    | pH                                                           | 5.5 – 9.0             |
| 2    | Temperature difference between incoming and outgoing streams | ≤ 5 °C                |
| 3    | Oil and grease                                               | 10                    |
| 4    | Suspended solids                                             | 100                   |
| 5    | BOD (3 days at 27°C)                                         | 30                    |
| 6    | COD                                                          | 250                   |
| 7    | Lead as Pb                                                   | 0.1                   |
| 8    | Hexavalent chromium (as Cr + 6)                              | 0.1                   |
| 9    | Total Chromium                                               | 2                     |
| 10   | Iron as Fe                                                   | 3                     |

### Noise Pollution

The plant can cause noise pollution due to operation of machineries, equipment and the operation of D.G sets.

The Consultants suggest the following noise pollution control measures:

- Machineries that are the major source of noise shall be properly maintained as per the maintenance schedule.
- Diesel engines shall be provided with acoustic enclosures and properly place on vibration isolators.
- Workers working in furnace area and crane operators shall be provided with earplugs and their duty hours shall be regulated.
- Green belt shall be developed around the premises, vacant spaces and around char dump area to attenuate noise and to prevent spread of fugitive emissions.

Noise generation from D.G set shall not exceed 85dB at a distance of 1 m from outline of equipment as per the CPCB guideline. However the plant shall conduct noise quality monitoring to check its compliance to ambient air quality w.r.t noise as indicated in Table .

Table : Permissible Limits of Noise Levels

| Area Code | Category of area/zone | Limits Day Time | Limits Night Time |
|-----------|-----------------------|-----------------|-------------------|
| A         | Industrial Area       | 75              | 70                |
| B         | Commercial Area       | 65              | 55                |
| C         | Residential Area      | 55              | 45                |
| D         | Science Zone          | 50              | 40                |

### Solid Waste Generation

The plant generates solid waste due to the following operations -

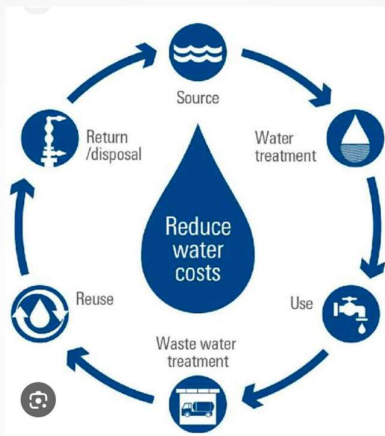
- Steel scrap during production of seamless tubes
- Solid waste from canteen, garbage and food waste.

The Consultants suggest the following measures to manage solid waste generated from the plant –

- Steel scrap generated shall be sent for recycling
- Solid waste generated shall be composted and used as manure for greenbelt.

## Utilities:

### Utilities



### Electricity:

The Consultants estimate that the power requirement will be about 35 MW for the entire plant. The same will be sourced from the Power grid.

| Sr.N o. | DESCRIPTION                              | CAPACITY      |
|---------|------------------------------------------|---------------|
| 01      | TOTAL PLANT CONNECTED LOAD               | 35 MVA        |
| 02      | PLANT CONTRACT DEMAND                    | 16 MVA        |
| 03      | PLANT REQUIRED INCOMING VOLTAGE          | 132 KV        |
| 04      | ESTIMATED AVG. MONTHLY POWER CONSUMPTION | 60 LAKH UNITS |

### Water:

The Consultants estimate that the Project will have a water requirement of about 240,000 Cu m annually.

Water will be needed for the following applications -

- Stage-wise cooling of tubes during the process mechanism
- Hydrostatic Testing
- Drinking
- Housekeeping
- Beautification

Based on discussion with the Management of the Company and best industry practice, the Consultants recommend –

- A large water body needs to be created within the site, which will also be used for holding runoff water during the monsoon
- Investments will have to be made in treatment facilities to recycle wastewater.

### Compressed Air:

The Consultants estimate that the Project will have a compressed air requirement of about  $25 \times 10^5 \text{ Nm}^3$  annually.

Compressed air will be required for supporting the combustion process, cooling process, and removing foreign particles from hollow tubes by dry blowing.

## Human Resources



### Organization chart:

The Project will operate for three shifts a day for about 330 days in a year. The Consultants estimate the requirement of 395 persons (direct and indirect) to operate the facility, as indicated in the Table below.

- The manpower estimated includes -
  - Plant and Corporate Management
  - Plant Engineers
  - Sales Engineers
- Technical and Non-technical staff in –
  - Operations & Maintenance
  - Utilities
  - Material handling
  - Administration
  - Sales and Marketing
  - Finance and Accounts
  - Procurement and Vendor Development
- Skilled Workmen in –
  - Operations & Maintenance
  - Utilities
  - Material handling
  - Stores
  - Administration
- Unskilled Workmen in –
  - Operations & Maintenance
  - Utilities
  - Material handling
  - Stores
  - Security

#### Manpower Details

| Description         | Numbers |
|---------------------|---------|
| Plant Head          | 1       |
| Managerial          | 12      |
| Executives          | 15      |
| Shift in charge     | 30      |
| Engineers           | 30      |
| Supervisory Staffs  | 15      |
| Operator            | 60      |
| Chief Metallurgist  | 1       |
| Metallurgist        | 2       |
| Lab Analyst         | 4       |
| Crane Operators     | 18      |
| Security Officer    | 3       |
| Security guards     | 16      |
| Helper              | 150     |
| Others- Electrician | 18      |
| Fitter / Welder     | 20      |
| Total               | 395     |

## Area required:

Total Estimated Land required for the plant will be Apprx.50 Acre

Break-up for Area wise is shown in below Table

| SR NO. | DESCRIPTION             | REQUIRED AREA IN( Sq.Mtr.) |
|--------|-------------------------|----------------------------|
| 01     | HOT MILL                | 28000                      |
| 02     | OCTG/API                | 22500                      |
| 03     | FINISHING               | 28850                      |
| 04     | RAW MATERIAL STORAGE    | 10000                      |
| 05     | FINISHED GOODS STORAGE  | 12000                      |
| 06     | ROAD AREA               | 32000                      |
| 07     | ADMINISTRATIVE BUILDING | 750 (3 FLOORS)             |
| 08     | CANTEEN                 | 200 (2 FLOORS)             |
| 09     | PARKING/GARDENING       | 12000                      |
| 10     | WORKER GUEST HOUSE      | 1000                       |
| 11     | ENGINEERS GUEST HOUSE   | 1000                       |
| 12     | UTILITIES               | 5000                       |
| 13     | ELECTRICAL              | 5000                       |
| 14     | STORES                  | 5000                       |
| 15     | FUTURE EXPANSION        | 40000                      |
|        |                         |                            |
| Total  |                         | 203300 (Apprx. 50 ACRE)    |

## Product specification



Seamless tubes are available in wide variety of wall thicknesses and diameters. Based on their physical and chemical properties they are preferred for a number of applications. Some of the important features of seamless tubes are:

- High Strength
- Corrosion Resistance
- Durability
- Long Life
- Fine Finish
- Recommended by all safety legislation in the world as a superior product for transportation of gas, fuel, pressure

Various standards of seamless tubes are available based on the type of industry to be catered to. Some of the prevalent standards are API, IS, ASTM, SAE, DIN and BS. The application of these standards as per the industry is listed in below table

| STANDARDS | APPLICABLE INDUSTRY                                                                                                                                                              |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API       | Oil & Gas                                                                                                                                                                        |
| IS        | Oil & Gas, Automotive, Hydrocarbon Processing Ind<br>Hydraulic Cylinder, Boiler, Heat Exchanger, Super Heater & Condenser, Railways & Mechanical Structure & General Engineering |
| ASTM      | Automotive, Hydrocarbon Processing Ind., Boiler, Heat Exchanger, Super heater & Condenser, Mechanical Structure & General Engineering                                            |
| SAE       | Automotive, Bearing Ind., Hydraulic Cylinder                                                                                                                                     |
| DIN       | Automotive, Hydraulic Cylinder, Boiler, Heat Exchanger, Super heater & Condenser, Mechanical Structure & General Engineering                                                     |
| BS        | Automotive, Hydrocarbon Processing Ind., Boiler, Heat Exchanger, Super heater & Condenser, Railways, Mechanical Structure & General Engineering                                  |



## PROPOSED BUDGET:

### Plant and Machinery



Plant and machinery cost is estimated at 6256.5 million as indicated in Table .

The total plant and machinery cost has further been divided to Tube mill and the OCTG plant. This is primarily because the OCTG plant will start operation after the Tube mill plant

Equipment estimated cost as per industry data has been assumed.

#### Cost of Plant and Machinery

|    | Plant & Machinery                                                                   | Total Value (₹) million | Remarks          |
|----|-------------------------------------------------------------------------------------|-------------------------|------------------|
| 1  | HOT LINE EQUIPMENT (MACHINERY) + RHF (IMPORTED)                                     | 2722.73                 | estimated cost   |
| 2  | FINISHING LINE EQUIPMENTS 1                                                         | 252.96                  | estimated cost   |
| 3  | FINISHING LINE EQUIPMENTS 2                                                         | 260.40                  | estimated cost   |
| 4  | OCTG LINE EQUIPMENTS+ TUBING EQUIPMENTS                                             | 1,765.60                | estimated cost   |
| 5  | TECHNOLOGICAL STRUCTURE                                                             | 508                     | estimated cost   |
| 6  | CRANES                                                                              | 77.95                   | estimated cost   |
| 7  | RHF Refractory cost + construction cost+ chimney                                    | 93.0                    | estimated cost   |
| 8  | SCALE PIT                                                                           | 10.0                    | estimated cost   |
| 9  | WATER PIPE LINES                                                                    | 18.0                    | estimated cost   |
| 10 | WATER TREATMENT PLANT                                                               | 20.5                    | as per quotation |
| 11 | Heat Treatment Furnace for Boiler pipe & Gas Pipe Line (incl instrument) (imported) | 174.40                  | estimated cost   |
| 12 | NITROGEN STATION                                                                    | 5.35                    | as per quotation |
| 13 | OCEAN FRIEGHT & LANDING & Insurance @10% of imported cost                           | 347.61                  |                  |
|    | <b>Total</b>                                                                        | <b>6256.5</b>           |                  |

## Project- Seamless Pipes & Tubes (Dia 60.3 – 278 mm size )

### Miscellaneous Fixed Asset:

The total MFA cost comes to ` 908.82 million. The breakup of the same is given in Table.

| Sl No.       | Miscellaneous Fixed assets                      | ` Million     | Remarks        |
|--------------|-------------------------------------------------|---------------|----------------|
| 1            | H.T. YARD 40 MVA + Transformer                  | 155.20        | estimated cost |
| 2            | ELEC. CABLES + PCC+MCC+Lighting                 | 97.60         | estimated cost |
| 3            | PUMP                                            | 3.22          | estimated cost |
| 5            | DEDUSTING SYSTEM with civil works               | 20.00         | estimated cost |
| 6            | Q. C.LAB. EQUIPMENTS                            | 12.00         | estimated cost |
| 7            | FIRE FIGHTER EQUIPMENTS                         | 5.00          | estimated cost |
| 8            | WEIGHING BRIDGE                                 | 3.75          | estimated cost |
| 9            | COMPRESSOR + Airline pipes                      | 17.50         | estimated cost |
| 10           | WELDING MACHINE + GAS CUTTING SET               | 5.00          | estimated cost |
| 11           | Tool Room                                       | 120.00        | estimated cost |
| 12           | PRODUCTION TOOLING(Imported) with ocean freight | 254.60        | estimated cost |
| 13           | MECH. & ELEC. MAINT. SPARE FOR TWO YEARS        | 100.00        | estimated cost |
| 14           | D. G. GENERATOR SET(1000Kw)                     | 8.55          | estimated cost |
| 15           | Furniture Fixtures                              | 7.00          | estimated cost |
| 16           | Computers/ Laptops                              | 10.00         | estimated cost |
| 17           | Coal Gassifier                                  | 90.00         | estimated cost |
| <b>Total</b> |                                                 | <b>908.82</b> |                |

Miscellaneous Fixed Asset ( ` in million)

Total of plant and machinery and MFA cost comes to 65.26% of the Project cost.

## Project- Seamless Pipes & Tubes (Dia 60.3 – 278 mm size )

### Preliminary and Pre-operative Expenses:

Preliminary and Preoperative expenses have been estimated at ₹356.40 million. The details are given in Table .

Table 9.8: Preliminary and Pre- operative expenses (in ₹ million)

| Sl No.       | Preliminary Pre op Expenses                     | ₹ Million     |
|--------------|-------------------------------------------------|---------------|
| 1            | Cost of power connection                        | 72.00         |
| 2            | Establishment, Training, Salaries               | 50.00         |
| 3            | Travelling Expenses                             | 35.00         |
| 4            | Upfront Fees and Financial Consultancy Expenses | 45.00         |
| 5            | Insurance                                       | 35.00         |
| 6            | Fees to Statutory Authorities                   | 26.00         |
| 7            | Cost of trial run                               | 90.00         |
| 8            | Consultancy for Water Management Services       | 3.40          |
| <b>Total</b> |                                                 | <b>356.40</b> |

Source: MM Estimates

The Preliminary and Preoperative expenses add up to 3.47% of the total Project cost.

**Proposed Total Budget:**  
**7521.72 Million INR (87.5 Million USD) Apprx.**

## Capacity and Capacity Utilization:

The Project shall have a Seamless Tube manufacturing facility with Line Pipe, special grade Boiler Pipe and OCTG pipes. Details of capacity and capacity utilization are as given in Table. It is to be noted that out of the total production 3% there will be a burning loss which has been represented as process loss. Further 9.5% of the total saleable product will be scrap which will be sold at a lesser price.

| Description                   | Unit          | Production |
|-------------------------------|---------------|------------|
| No. of Operating Days         | Days          | 330        |
|                               |               |            |
| <b>Seamless Tube Capacity</b> | MTPA          | 240000     |
|                               |               |            |
| <b>Line Pipe Capacity</b>     | MTPD          | 232        |
| Line Pipe                     | MTPA          | 76800      |
| Capacity Utilization          | %             | 75%        |
|                               |               |            |
| <b>Boiler Pipe Capacity</b>   | MTPD          | 58         |
| Boiler Pipe                   | MTPA          | 19200      |
| Capacity Utilization          | %             | 60%        |
|                               |               |            |
| <b>OCTG</b>                   |               |            |
| Casing                        | MTPD          | 290        |
| Tubing                        | MTPD          | 102        |
| Coupling                      | MTPD          | 44         |
| Total OCTG                    | MTPD          | 436        |
| Total OCTG Capacity           | MTPA          | 144000     |
| Capacity Utilization          | %             | 70%        |
| Process Loss                  | %             | 3%         |
| Scrap Grade 1                 | % of input RM | 30 %       |
| Scrap Grade 2                 | % of input RM | 30 %       |
|                               |               |            |
|                               |               |            |

### Revenue:

The selling price considered for Line Pipe, Boiler Pipe and OCTG is given in Table . The Consultants have taken conservative selling prices, which are based on the historical trend of price movements observed in the last six months of 2024-25. Prices of steel are highly volatile and hence a sensitivity analysis has been undertaken to look at viability even at adverse situation.

Export prices have been considered as 5% higher than that of domestic prices for line pipe and special grade boiler pipe. For OCTG the last year average prices in US market as reported by Metal Bulletin has been considered.

| Products        | Types of Price | Tentative Selling Price (₹ / MT) |
|-----------------|----------------|----------------------------------|
| Line Pipe -     | Market Price   | 85,000                           |
| Boiler Pipe -   | Market Price   | 105,000                          |
| OCTG -          | Market Price   | 11,0000                          |
| Scrap - Grade 1 | Market Price   | 39,000                           |
| Scrap - Grade 2 | Market Price   | 29,000                           |



## Spares and Consumables

The spares and consumable cost has been taken as per industry standards. Details of the same are given in Table .

Table : Specific Consumption for Line Pipe and OCTG with unit prices

| Description                                      | Unit      | Quantity | Rates<br>(`./Unit) |
|--------------------------------------------------|-----------|----------|--------------------|
| <b>Raw Material</b>                              |           |          |                    |
| Billet                                           | MT/MT     | 1.10     | 55,000-65000       |
| Natural Gas                                      | Nm3/ MT   | 12       | 36                 |
| <b>Utilities</b>                                 |           |          |                    |
| Water                                            | cu m/ MT  | 1.22     | 22.0               |
| Power                                            | units/ MT | 245.00^  | 5.50               |
| Compressed Air                                   | Nm3/ MT   | 7.50     | 0.50               |
| Coal                                             | Kg/ MT    | 222.0    | 6.00               |
| Roll & Tooling                                   | ` / MT    |          | 7                  |
| Refractory                                       | ` / MT    |          | 2                  |
| Lubricating Oil                                  | ` / MT    |          | 1                  |
| Graphite Powder                                  | ` / MT    |          | 4                  |
| Borax Powder                                     | ` / MT    |          | 1                  |
| Conversion Cost – Heat Treatment (only for OCTG) | ` / MT    |          | 3,000              |

- Power consumption has been assumed to be 245 units/MT in the first year and subsequently reduced by 10% to reach 200 units/ MT once the plant is fully operational.

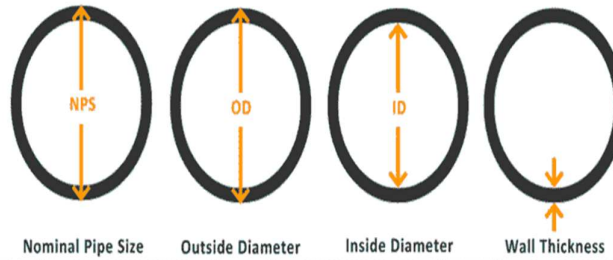
The heat treatment cost is only applicable to OCTG pipes and hence a conversion cost of ` 3,000 per MT has been taken.

## API 5L Pipes Mechanical Properties

| API 5L Sour Service Pipe Tensile Tests Results        |                                                       |             |                                                    |              |                                      |                                                |                                                       |
|-------------------------------------------------------|-------------------------------------------------------|-------------|----------------------------------------------------|--------------|--------------------------------------|------------------------------------------------|-------------------------------------------------------|
| Pipe steel grade                                      | Yield strength <sup>a</sup><br>$R_{10.5}$<br>MPa(psi) |             | Tensile strength <sup>a</sup><br>$R_m$<br>MPa(psi) |              | Ratio <sup>b</sup><br>$R_{10.5}/R_m$ | Elongation (on<br>50 mm or 2 in)<br>$A_1$<br>% | Tensile<br>strength <sup>c</sup><br>$R_m$<br>MPa(psi) |
|                                                       | minimum                                               | maximum     | minimum                                            | maximum      | maximum                              | minimum                                        | minimum                                               |
| L245NS or BNS<br>L245QS or BQS<br>L245MS or BMS       | 245 (35500)                                           | 450 (65300) | 415 (60200)                                        | 655 (95000)  | 0.93                                 | e                                              | 415 (60200)                                           |
| L290NS or X42NS<br>L290QS or X42QS<br>L290MS or X42MS | 290 (42100)                                           | 495 (71800) | 415 (60200)                                        | 655 (95000)  | 0.93                                 | e                                              | 415 (60200)                                           |
| L320NS or X46NS<br>L320QS or X46QS<br>L320MS or X46MS | 320 (46400)                                           | 525 (76100) | 435 (63100)                                        | 655 (95000)  | 0.93                                 | e                                              | 435 (63100)                                           |
| L360NS or X52NS<br>L360QS or X52QS<br>L360MS or X52MS | 360 (52200)                                           | 530 (76900) | 460 (66700)                                        | 760 (110200) | 0.93                                 | e                                              | 460 (66700)                                           |
| L390QS or X56QS<br>L390MS or X56MS                    | 390 (56600)                                           | 545 (79000) | 490 (71100)                                        | 760 (110200) | 0.93                                 | e                                              | 490 (71100)                                           |
| L415QS or X60QS<br>L415MS or X60MS                    | 415 (60200)                                           | 565 (81900) | 520 (75400)                                        | 760 (110200) | 0.93                                 | e                                              | 520 (75400)                                           |
| L450QS or X65QS<br>L450MS or X65MS                    | 450 (65300)                                           | 600 (87000) | 535 (77600)                                        | 760 (110200) | 0.93                                 | e                                              | 535 (77600)                                           |
| L485QS or X70QS<br>L485MS or X70MS                    | 485 (70300)                                           | 635 (92100) | 570 (82700)                                        | 760 (110200) | 0.93                                 | e                                              | 570 (82700)                                           |

# Project- Seamless Pipes & Tubes (Dia 60.3 – 278 mm size )

API 5L Pipe size chart



| NOMINAL Pipe Size |        | OUTSIDE DIAMETER | NOMINAL WALL THICKNESS |       |       |      |       |       |       |       |         |         |         |         |       |
|-------------------|--------|------------------|------------------------|-------|-------|------|-------|-------|-------|-------|---------|---------|---------|---------|-------|
| DN                | OD     | ASME             | SCH10                  | SCH20 | SCH30 | STD  | SCH40 | SCH60 | XS    | SCH80 | SCH 100 | SCH 120 | SCH 140 | SCH 160 | XXS   |
| 15                | 1/2"   | 21.3             | 2.11                   |       | 2.41  | 2.77 | 2.77  |       | 3.73  | 3.73  |         |         |         | 4.78    | 7.47  |
| 20                | 3/4"   | 26.7             | 2.11                   |       | 2.41  | 2.87 | 2.87  |       | 3.91  | 3.91  |         |         |         | 5.56    | 7.82  |
| 25                | 1"     | 33.4             | 2.77                   |       | 2.9   | 3.38 | 3.38  |       | 4.55  | 4.55  |         |         |         | 6.35    | 9.09  |
| 32                | 1-1/4" | 42.2             | 2.77                   |       | 2.97  | 3.56 | 3.56  |       | 4.85  | 4.85  |         |         |         | 6.35    | 9.7   |
| 40                | 1-1/2" | 48.3             | 2.77                   |       | 3.128 | 3.68 | 3.68  |       | 5.08  | 5.08  |         |         |         | 7.14    | 10.15 |
| 50                | 2"     | 60.3             | 2.77                   |       | 3.18  | 3.91 | 3.91  |       | 5.54  | 5.54  |         |         |         | 8.74    | 11.07 |
| 65                | 2-1/2" | 73               | 3.05                   |       | 4.78  | 5.16 | 5.16  |       | 7.01  | 7.01  |         |         |         | 9.53    | 14.02 |
| 80                | 3"     | 88.9             | 3.05                   |       | 4.78  | 5.49 | 5.49  |       | 7.62  | 7.62  |         |         |         | 11.13   | 15.25 |
| 90                | 3-1/2" | 101.6            | 3.05                   |       | 4.78  | 5.74 | 5.74  |       | 8.08  | 8.08  |         |         |         |         |       |
| 100               | 4"     | 114.3            | 3.05                   |       | 4.78  | 6.02 | 6.02  |       | 8.56  | 8.56  |         | 11.3    |         | 13.49   | 17.12 |
| 125               | 5"     | 141.3            | 3.4                    |       |       | 6.55 | 6.55  |       | 9.53  | 9.53  |         | 12.7    |         | 15.88   | 19.05 |
| 150               | 6"     | 168.3            | 3.4                    |       |       | 7.11 | 7.11  |       | 10.97 | 10.97 |         | 14.27   |         | 18.26   | 21.95 |
| 200               | 7"     | 219.1            | 3.76                   |       | 7.04  | 8.18 | 8.18  | 10.31 | 12.7  | 12.7  | 15.09   | 18.26   | 20.62   | 23.01   | 22.23 |
| 250               | 8"     | 273              | 3.76                   | 6.35  | 7.8   | 9.27 | 9.27  | 12.7  | 12.7  | 15.09 | 18.26   | 21.44   | 25.4    | 28.58   | 25.4  |
| 300               | 10"    | 232.8            | 4.19                   | 6.35  | 8.38  | 9.53 | 10.31 | 14.27 | 12.7  | 17.48 | 21.44   | 25.4    | 28.58   | 33.32   | 25.4  |
| 350               | 12"    | 355.6            | 4.57                   | 6.35  | 9.53  | 9.53 | 11.13 | 15.09 | 12.7  | 19.05 | 23.83   | 27.79   | 31.75   | 35.71   |       |
| 400               | 14"    | 406.4            | 6.35                   | 7.92  | 9.53  | 9.53 | 12.7  | 16.66 | 12.7  | 21.44 | 26.19   | 30.96   | 36.53   | 40.19   |       |
| 450               | 16"    | 457.2            | 6.35                   | 7.92  | 11.13 | 9.53 | 14.27 | 19.05 | 12.7  | 23.83 | 39.36   | 34.93   | 39.67   | 45.24   |       |
| 500               | 20"    | 508              | 6.35                   | 9.53  | 12.7  | 9.53 | 15.09 | 20.62 | 12.7  | 26.19 | 32.54   | 38.1    | 44.45   | 50.01   |       |
| 550               | 22"    | 558.8            | 6.35                   | 9.53  | 12.7  | 9.53 |       | 22.23 | 12.7  | 28.58 | 34.93   | 41.28   | 47.63   | 53.98   |       |
| 600               | 24"    | 609.6            | 6.35                   | 9.53  | 14.7  | 9.53 | 17.48 | 24.61 | 12.7  | 30.96 | 38.89   | 46.02   | 52.37   | 59.54   |       |
| 650               | 26"    | 660.4            | 7.92                   | 12.7  |       | 9.53 |       |       | 12.7  |       |         |         |         |         |       |
| 700               | 28"    | 711.2            | 7.92                   | 12.7  | 15.88 | 9.53 |       |       | 12.7  |       |         |         |         |         |       |
| 750               | 30"    | 762              | 7.92                   | 12.7  | 15.88 | 9.53 |       |       | 12.7  |       |         |         |         |         |       |

## API 5L Chemical compositions

| API 5L Sour Service Pipe with wall thickness ≤ 25.0 mm (0.984 in) |                                                     |      |                 |       |                    |      |      |      |                      |                                |                   |
|-------------------------------------------------------------------|-----------------------------------------------------|------|-----------------|-------|--------------------|------|------|------|----------------------|--------------------------------|-------------------|
| API 5L Grade                                                      | Mass fraction, based upon heat and product analyses |      |                 |       |                    |      |      |      |                      | Carbon equivalent<br>% maximum |                   |
|                                                                   | % maximum                                           |      |                 |       |                    |      |      |      |                      |                                |                   |
|                                                                   | C <sup>b</sup>                                      | Si   | Mn <sup>b</sup> | P     | S                  | V    | Nb   | Ti   | other <sup>c,d</sup> | CE <sub>IIW</sub>              | CE <sub>PCM</sub> |
| Seamless and welded pipes                                         |                                                     |      |                 |       |                    |      |      |      |                      |                                |                   |
| L245NS or BNS                                                     | 0.14                                                | 0.40 | 1.35            | 0.020 | 0.003 <sup>e</sup> |      |      | 0.04 | g                    | 0.36                           | 0.19 <sup>h</sup> |
| L290NS or X42NS                                                   | 0.14                                                | 0.40 | 1.35            | 0.020 | 0.003 <sup>e</sup> | 0.05 | 0.05 | 0.04 | -                    | 0.36                           | 0.19 <sup>h</sup> |
| L320NS or X46NS                                                   | 0.14                                                | 0.40 | 1.40            | 0.020 | 0.003 <sup>e</sup> | 0.07 | 0.05 | 0.04 | g                    | 0.38                           | 0.20 <sup>h</sup> |
| L360NS or X52NS                                                   | 0.16                                                | 0.45 | 1.65            | 0.020 | 0.003 <sup>e</sup> | 0.10 | 0.05 | 0.04 | g                    | 0.43                           | 0.22 <sup>h</sup> |
| L245QS or XQBS                                                    | 0.14                                                | 0.40 | 1.35            | 0.020 | 0.003 <sup>e</sup> | 0.04 | 0.04 | 0.04 | -                    | 0.34                           | 0.19 <sup>h</sup> |
| L290QS or X42QS                                                   | 0.14                                                | 0.40 | 1.35            | 0.020 | 0.003 <sup>e</sup> | 0.04 | 0.04 | 0.04 | -                    | 0.34                           | 0.19 <sup>h</sup> |
| L320QS or X46QS                                                   | 0.15                                                | 0.45 | 1.40            | 0.020 | 0.003 <sup>e</sup> | 0.05 | 0.05 | 0.04 | -                    | 0.36                           | 0.20 <sup>h</sup> |
| L360QS or X52QS                                                   | 0.16                                                | 0.45 | 1.65            | 0.020 | 0.003 <sup>e</sup> | 0.07 | 0.05 | 0.04 | g                    | 0.39                           | 0.20 <sup>h</sup> |
| L390QS or X56QS                                                   | 0.16                                                | 0.45 | 1.65            | 0.020 | 0.003 <sup>e</sup> | 0.07 | 0.05 | 0.04 | g                    | 0.40                           | 0.21 <sup>h</sup> |
| L415QS or X60QS                                                   | 0.16                                                | 0.45 | 1.65            | 0.020 | 0.003 <sup>e</sup> | 0.08 | 0.05 | 0.04 | g,i,k                | 0.41                           | 0.22 <sup>h</sup> |
| L450QS or X65QS                                                   | 0.16                                                | 0.45 | 1.65            | 0.020 | 0.003 <sup>e</sup> | 0.09 | 0.05 | 0.06 | g,i,k                | 0.42                           | 0.22 <sup>h</sup> |
| L485QS or X70QS                                                   | 0.16                                                | 0.45 | 1.65            | 0.020 | 0.003 <sup>e</sup> | 0.09 | 0.05 | 0.06 | g,i,k                | 0.42                           | 0.22 <sup>h</sup> |
| Welded pipe                                                       |                                                     |      |                 |       |                    |      |      |      |                      |                                |                   |
| L245MS or BMS                                                     | 0.10                                                | 0.40 | 1.25            | 0.020 | 0.002 <sup>e</sup> | 0.04 | 0.04 | 0.04 | -                    | -                              | 0.19              |
| L290MS or X42MS                                                   | 0.10                                                | 0.40 | 1.25            | 0.020 | 0.002 <sup>e</sup> | 0.04 | 0.04 | 0.04 | -                    | -                              | 0.19              |
| L320MS or X46MS                                                   | 0.10                                                | 0.45 | 1.35            | 0.020 | 0.002 <sup>e</sup> | 0.05 | 0.05 | 0.04 | -                    | -                              | 0.20              |
| L360MS or X52MS                                                   | 0.10                                                | 0.45 | 1.45            | 0.020 | 0.002 <sup>e</sup> | 0.05 | 0.06 | 0.04 | -                    | -                              | 0.20              |
| L390MS or X56MS                                                   | 0.10                                                | 0.45 | 1.45            | 0.020 | 0.002 <sup>e</sup> | 0.06 | 0.08 | 0.04 | g                    | -                              | 0.21              |
| L415MS or X60MS                                                   | 0.10                                                | 0.45 | 1.45            | 0.020 | 0.002 <sup>e</sup> | 0.08 | 0.08 | 0.06 | g,i                  | -                              | 0.21              |
| L450MS or X65MS                                                   | 0.10                                                | 0.45 | 1.60            | 0.020 | 0.002 <sup>e</sup> | 0.10 | 0.08 | 0.06 | g,i,k                | -                              | 0.22              |
| L485MS or X70MS                                                   | 0.10                                                | 0.45 | 1.60            | 0.020 | 0.002 <sup>e</sup> | 0.10 | 0.08 | 0.06 | g,i,k                | -                              | 0.22              |

a. If C > 0.12%, CE<sub>IIW</sub> limits shall be applied; If C ≤ 0.12%, CE<sub>PCM</sub> shall be applied.

b. For each reduce of 0.01% for maximum C, an increase of 0.05% maximum Mn is permissible, up to a maximum of 0.20%.

c. Al ≤ 0.060%; N ≤ 0.012%; Al/N ≥ 2:1 (titanium-killed or titanium-treated steel not applicable);

Cu ≤ 0.35% (Cu ≤ 0.10% if agreed); Ni ≤ 0.30%; Cr ≤ 0.30%; Mo ≤ 0.15%; B ≤ 0.0005%.

d. For seamless and welded pipes, Ca ≤ 0.006%; For welded pipe if Ca is added by intention, unless agreed, Ca/S ≥ 1.5 in case S > 0.0015%.

e. For SMLS pipe maximum limit for S could be increased to ≤ 0.008%, and in case welded if agreed to ≤ 0.006%. For higher S content in welded pipe, lower Ca/S ratios maybe agreed.

f. Nb + V ≤ 0.06%, unless otherwise agreed.

g. Nb + V + Ti ≤ 0.15%.

h. In case seamless pipe, listed CE<sub>PCM</sub> value could be increased by 0.03.

i. Mo ≤ 0.35% in case agreed.

j. Cr ≤ 0.45% in case agreed.

k. Cr ≤ 0.45% and Ni ≤ 0.50% in case agreed.



# Project- Seamless Pipes & Tubes (Dia 60.3 – 278 mm size )

API 5 CT Dimensions and Weight Chart:

| Lables |        |       |        | Outside<br>dia. | Nominal linear masses |                  |                    | Wall<br>thickness | Inside<br>dia. | Mass gain or loss due to end finishing lb |                |         |                      |                   |
|--------|--------|-------|--------|-----------------|-----------------------|------------------|--------------------|-------------------|----------------|-------------------------------------------|----------------|---------|----------------------|-------------------|
|        |        |       |        |                 |                       |                  |                    |                   |                | Plain end                                 | External upset |         |                      | Integral<br>joint |
| 1      | 2      |       |        |                 | Non-<br>upset         | Ext.upset<br>T&C | Ontergral<br>joint |                   |                |                                           | Non<br>upset   | Regular | Special<br>clearance |                   |
| NU T&C | EU T&C | U     | D (in) | lb/ft           | lb/ft                 | lb/ft            | t (in)             | d (in)            |                |                                           |                |         |                      |                   |
| 1      | 2      | 3     | 4      | 5               | 6                     | 7                | 8                  | 9                 | 10             | 11                                        | 12             | 13      | 14                   | 15                |
| 1.050  | 1.14   | 1.20  |        | 1.050           | 1.14                  | 1.20             |                    | 0.113             | 0.824          | 1.13                                      | 0.20           | 1.40    |                      |                   |
| 1.050  | 1.48   | 1.54  |        | 1.050           | 1.48                  | 1.54             |                    | 0.154             | 0.742          | 1.48                                      |                | 1.32    |                      |                   |
| 1.315  | 1.70   | 1.80  | 1.72   | 1.315           | 1.70                  | 1.80             | 1.72               | 0.133             | 1.049          | 1.68                                      | 0.40           | 1.40    |                      | 0.20              |
| 1.315  | 2.19   | 2.24  |        | 1.315           | 2.19                  | 2.24             |                    | 0.179             | 0.957          | 2.17                                      |                | 1.35    |                      |                   |
| 1.660  | 2.09   |       | 2.10   | 1.660           |                       |                  | 2.10               | 0.125             | 1.410          | 2.05                                      |                |         |                      | 0.20              |
| 1.660  | 2.30   | 2.40  | 2.33   | 1.660           | 2.30                  | 2.40             | 2.33               | 0.140             | 1.380          | 2.27                                      | 0.80           | 1.60    |                      | 0.20              |
| 1.660  | 3.03   | 3.07  |        | 1.660           | 3.03                  | 3.07             |                    | 0.191             | 1.278          | 3.00                                      |                | 1.50    |                      |                   |
| 1.900  | 2.40   |       | 2.40   | 1.990           |                       |                  | 2.40               | 0.125             | 1.650          | 2.37                                      |                |         |                      | 0.20              |
| 1.900  | 2.75   | 2.90  | 2.76   | 1.990           | 2.75                  | 2.90             | 2.76               | 0.145             | 1.610          | 2.72                                      | 0.60           | 2.00    |                      | 0.20              |
| 1.900  | 3.65   | 3.73  |        | 1.990           | 3.65                  | 3.73             |                    | 0.200             | 1.500          | 3.63                                      |                | 2.03    |                      |                   |
| 1.900  | 4.42   |       |        | 1.990           | 4.42                  |                  |                    | 0.250             | 1.400          | 4.41                                      |                |         |                      |                   |
| 1.900  | 5.15   |       |        | 1.990           | 5.15                  |                  |                    | 0.300             | 1.300          | 5.13                                      |                |         |                      |                   |
| 2.063  | 3.24   |       |        | 2.063           |                       |                  | 3.25               | 0.156             | 1.751          | 3.18                                      |                |         |                      | 0.20              |
| 2.063  | 4.50   |       |        | 2.063           |                       |                  |                    | 0.225             | 1.613          | 4.42                                      |                |         |                      |                   |
| 2 3/8  | 4.00   |       |        | 2.375           | 4.00                  |                  |                    | 0.167             | 2.041          | 3.94                                      | 1.60           |         |                      |                   |
| 2 3/8  | 4.60   | 4.70  |        | 2.375           | 4.60                  | 4.70             |                    | 0.190             | 1.995          | 4.44                                      | 1.60           | 4.00    | 2.96                 |                   |
| 2 3/8  | 5.80   | 5.95  |        | 2.375           | 5.80                  | 5.95             |                    | 0.254             | 1.867          | 5.76                                      | 1.40           | 3.60    | 2.56                 |                   |
| 2 3/8  | 6.60   |       |        | 2.375           | 6.60                  |                  |                    | 0.295             | 1.785          | 6.56                                      |                |         |                      |                   |
| 2 3/8  | 7.35   | 7.45  |        | 2.375           | 7.35                  | 7.45             |                    | 0.336             | 1.703          | 7.32                                      |                |         |                      |                   |
| 2 7/8  | 6.40   | 6.50  |        | 2.875           | 6.40                  | 6.50             |                    | 0.217             | 2.441          | 6.17                                      | 3.20           | 5.60    | 3.76                 |                   |
| 2 7/8  | 7.80   | 7.90  |        | 2.875           | 7.80                  | 7.90             |                    | 0.276             | 2.323          | 7.67                                      | 0.80           | 5.80    | 3.92                 |                   |
| 2 7/8  | 8.60   | 8.70  |        | 2.875           | 8.60                  | 8.70             |                    | 0.308             | 2.259          | 8.45                                      | 2.60           | 5.00    | 3.16                 |                   |
| 2 7/8  | 9.35   | 9.45  |        | 2.875           | 9.35                  | 9.45             |                    | 0.340             | 2.195          | 9.21                                      |                |         |                      |                   |
| 2 7/8  | 10.50  |       |        | 2.875           | 10.50                 |                  |                    | 0.392             | 2.091          | 10.40                                     |                |         |                      |                   |
| 2 7/8  | 11.50  |       |        | 2.875           | 11.50                 |                  |                    | 0.440             | 1.995          | 11.45                                     |                |         |                      |                   |
| 3 1/2  | 7.70   |       |        | 3.500           | 7.70                  |                  |                    | 0.216             | 3.068          | 7.58                                      | 5.40           |         |                      |                   |
| 3 1/2  | 9.20   | 9.30  |        | 3.500           | 9.20                  | 9.30             |                    | 0.254             | 2.992          | 8.81                                      | 5.00           | 9.20    | 5.40                 |                   |
| 3 1/2  | 10.20  |       |        | 3.500           | 10.20                 |                  |                    | 0.289             | 2.992          | 9.92                                      | 4.80           |         |                      |                   |
| 3 1/2  | 12.70  | 12.95 |        | 3.500           | 12.70                 | 12.95            |                    | 0.375             | 2.750          | 12.53                                     | 4.00           | 8.20    | 4.40                 |                   |
| 3 1/2  | 14.30  |       |        | 3.500           | 14.30                 |                  |                    | 0.430             | 2.640          | 14.11                                     |                |         |                      |                   |
| 3 1/2  | 15.50  |       |        | 3.500           | 15.50                 |                  |                    | 0.476             | 2.548          | 15.39                                     |                |         |                      |                   |
| 3 1/2  | 17.00  |       |        | 3.500           | 17.00                 |                  |                    | 0.530             | 2.440          | 16.83                                     |                |         |                      |                   |
| 4      | 9.50   |       |        | 4.000           | 9.50                  |                  |                    | 0.226             | 3.548          | 9.12                                      | 6.20           |         |                      |                   |
| 4      | 10.70  | 11.00 |        | 4.000           |                       | 11.00            |                    | 0.262             | 3.476          | 10.47                                     |                | 10.60   |                      |                   |
| 4      | 13.20  |       |        | 4.000           | 13.20                 |                  |                    | 0.330             | 3.340          | 12.95                                     |                |         |                      |                   |
| 4      | 16.10  |       |        | 4.000           | 16.10                 |                  |                    | 0.415             | 3.170          | 15.90                                     |                |         |                      |                   |
| 4      | 18.90  |       |        | 4.000           | 18.90                 |                  |                    | 0.500             | 3.000          | 18.71                                     |                |         |                      |                   |
| 4      | 22.20  |       |        | 4.000           | 22.20                 |                  |                    | 0.610             | 2.780          | 22.11                                     |                |         |                      |                   |
| 4 1/2  | 12.60  | 12.75 |        | 4.500           | 12.60                 | 12.75            |                    | 0.271             | 3.958          | 12.25                                     | 6.00           | 13.20   |                      |                   |
| 4 1/2  | 15.20  |       |        | 4.500           | 15.20                 |                  |                    | 0.337             | 3.826          | 15.00                                     |                |         |                      |                   |
| 4 1/2  | 17.00  |       |        | 4.500           | 17.00                 |                  |                    | 0.380             | 3.740          | 16.77                                     |                |         |                      |                   |
| 4 1/2  | 18.90  |       |        | 4.500           | 18.90                 |                  |                    | 0.430             | 3.640          | 18.71                                     |                |         |                      |                   |
| 4 1/2  | 21.50  |       |        | 4.500           | 21.50                 |                  |                    | 0.500             | 3.500          | 21.38                                     |                |         |                      |                   |
| 4 1/2  | 23.70  |       |        | 4.500           | 23.70                 |                  |                    | 0.560             | 3.380          | 23.59                                     |                |         |                      |                   |
| 4 1/2  | 26.10  |       |        | 4.500           | 26.10                 |                  |                    | 0.630             | 3.240          | 26.06                                     |                |         |                      |                   |

# Project- Seamless Pipes & Tubes (Dia 60.3 – 278 mm size )

**Steel Casing Pipe Dimensions and Weight Chart:**

| Lables |       | Outside diameter | Nominal linear mass T&C | Wall thickness | Inside diameter | Drift diameter | Calculated mass C |                                             |       |                 |       |
|--------|-------|------------------|-------------------------|----------------|-----------------|----------------|-------------------|---------------------------------------------|-------|-----------------|-------|
|        |       |                  |                         |                |                 |                | Plain end         | Mass gain or loss due to end finishing (lb) |       |                 |       |
|        |       |                  |                         |                |                 |                |                   | Round thread                                |       | Buttress thread |       |
| 1      | 2.00  | D (in)           | lb/ft                   | t (in)         | d (in)          | in             | lb/ft             | Short                                       | Long  | RC              | SCC   |
| 1      | 2     | 3                | 4                       | 5              | 6               | 7              | 8                 | 9                                           | 10    | 11              | 12    |
| 4 1/2  | 9.50  | 4.500            | 9.50                    | 0.205          | 4.090           | 3.965          | 9.41              | 4.20                                        |       |                 |       |
| 4 1/2  | 10.50 | 4.500            | 10.50                   | 0.224          | 4.052           | 3.927          | 10.24             | 3.80                                        |       | 5.00            | 2.56  |
| 4 1/2  | 11.60 | 4.500            | 11.60                   | 0.250          | 4.000           | 3.875          | 11.36             | 3.40                                        | 3.80  | 4.60            | 2.16  |
| 4 1/2  | 13.50 | 4.500            | 13.50                   | 0.290          | 3.920           | 3.795          | 13.05             |                                             | 3.20  | 4.00            | 1.56  |
| 4 1/2  | 15.10 | 4.500            | 15.10                   | 0.337          | 3.826           | 3.701          | 15.00             |                                             | 2.80  | 3.20            | 0.76  |
| 5      | 11.50 | 5.000            | 11.50                   | 0.220          | 4.560           | 4.435          | 11.24             | 5.40                                        |       |                 |       |
| 5      | 13.00 | 5.000            | 13.00                   | 0.253          | 4.494           | 4.369          | 12.84             | 4.80                                        | 5.80  | 6.60            | 2.42  |
| 5      | 15.00 | 5.000            | 15.00                   | 0.296          | 4.408           | 4.283          | 14.88             | 4.20                                        | 5.20  | 5.80            | 1.62  |
| 5      | 18.00 | 5.000            | 18.00                   | 0.362          | 4.276           | 4.151          | 17.95             |                                             | 4.20  | 4.40            | 0.22  |
| 5      | 21.40 | 5.000            | 21.40                   | 0.437          | 4.126           | 4.001          | 21.32             |                                             | 2.95  | 2.46            | -1.72 |
| 5      | 23.20 | 5.000            | 23.20                   | 0.478          | 4.044           | 3.919          | 23.11             |                                             | 2.30  | 2.05            | -2.09 |
| 5      | 24.10 | 5.000            | 24.10                   | 0.500          | 4.000           | 3.875          | 24.05             |                                             | 1.95  | 1.24            | -2.94 |
| 5 1/2  | 14.00 | 5.500            | 14.00                   | 0.244          | 5.015           | 4.887          | 13.71             | 5.40                                        |       |                 |       |
| 5 1/2  | 15.50 | 5.500            | 15.50                   | 0.275          | 4.950           | 4.825          | 15.36             | 4.80                                        | 5.80  | 6.40            | 2.10  |
| 5 1/2  | 17.00 | 5.500            | 17.00                   | 0.304          | 4.892           | 4.767          | 16.89             | 4.40                                        | 5.40  | 5.80            | 1.50  |
| 5 1/2  | 20.00 | 5.500            | 20.00                   | 0.361          | 4.778           | 4.653          | 19.83             |                                             | 4.40  | 4.60            | 0.30  |
| 5 1/2  | 23.00 | 5.500            | 23.00                   | 0.415          | 4.670           | 4.545          | 22.56             |                                             | 3.20  | 3.40            | 0.90  |
| 5 1/2  | 26.80 | 5.500            | 26.80                   | 0.500          | 4.500           | 4.375          | 26.72             |                                             |       |                 |       |
| 5 1/2  | 29.70 | 5.500            | 29.70                   | 0.562          | 4.376           | 4.251          | 29.67             |                                             |       |                 |       |
| 5 1/2  | 32.60 | 5.500            | 32.60                   | 0.625          | 4.250           | 4.125          | 32.57             |                                             |       |                 |       |
| 5 1/2  | 35.30 | 5.500            | 35.30                   | 0.687          | 4.126           | 4.001          | 35.35             |                                             |       |                 |       |
| 5 1/2  | 38.00 | 5.500            | 38.00                   | 0.750          | 4.000           | 3.875          | 38.08             |                                             |       |                 |       |
| 5 1/2  | 40.50 | 5.500            | 40.50                   | 0.812          | 3.876           | 3.751          | 40.69             |                                             |       |                 |       |
| 5 1/2  | 43.10 | 5.500            | 43.10                   | 0.875          | 3.750           | 3.625          | 43.26             |                                             |       |                 |       |
| 6 5/8  | 20.00 | 6.625            | 20.00                   | 0.288          | 6.049           | 5.924          | 19.51             | 11.00                                       | 13.60 | 14.40           | 2.38  |
| 6 5/8  | 24.00 | 6.625            | 24.00                   | 0.352          | 5.921           | 5.796          | 23.60             | 9.60                                        | 12.00 | 12.60           | 0.58  |
| 6 5/8  | 28.00 | 6.625            | 28.00                   | 0.417          | 5.791           | 5.666          | 27.67             |                                             | 10.20 | 10.60           | -1.42 |
| 6 5/8  | 32.00 | 6.625            | 32.00                   | 0.475          | 5.675           | 5.550          | 31.23             |                                             | 8.80  | 9.00            | -3.02 |
| 7      | 17.00 | 7.000            | 17.00                   | 0.231          | 6.538           | 6.413          | 16.72             | 10.00                                       |       |                 |       |
| 7      | 20.00 | 7.000            | 20.00                   | 0.272          | 6.456           | 6.331          | 19.56             | 9.40                                        |       |                 |       |
| 7      | 23.00 | 7.000            | 23.00                   | 0.317          | 6.366           | 6.250          | 22.65             | 8.00                                        | 10.40 | 11.00           | 1.60  |
| 7      | 23.00 | 7.000            | 23.00                   | 0.317          | 6.366           | 6.241          | 22.65             | 8.00                                        | 10.40 | 11.00           | 1.60  |
| 7      | 26.00 | 7.000            | 26.00                   | 0.362          | 6.276           | 6.151          | 25.69             | 7.20                                        | 9.40  | 9.60            | 0.20  |
| 7      | 29.00 | 7.000            | 29.00                   | 0.408          | 6.184           | 6.059          | 28.75             |                                             | 8.00  | 8.20            | 1.20  |
| 7      | 32.00 | 7.000            | 32.00                   | 0.453          | 6.094           | 6.000          | 31.70             |                                             | 6.60  | 6.80            | 2.60  |
| 7      | 32.00 | 7.000            | 32.00                   | 0.453          | 6.094           | 5.969          | 31.70             |                                             | 6.60  | 6.80            | 2.60  |
| 7      | 35.00 | 7.000            | 35.00                   | 0.498          | 6.004           | 5.879          | 34.61             |                                             | 5.60  | 5.60            | 3.80  |
| 7      | 38.00 | 7.000            | 38.00                   | 0.540          | 5.920           | 5.795          | 37.29             |                                             | 4.40  | 4.20            | 5.20  |
| 7      | 42.70 | 7.000            | 42.70                   | 0.625          | 5.750           | 5.625          | 42.59             |                                             |       |                 |       |
| 7      | 46.40 | 7.000            | 46.40                   | 0.687          | 5.625           | 5.500          | 46.36             |                                             |       |                 |       |
| 7      | 50.10 | 7.000            | 50.10                   | 0.750          | 5.500           | 5.375          | 50.11             |                                             |       |                 |       |
| 7      | 53.60 | 7.000            | 53.60                   | 0.812          | 5.376           | 5.251          | 53.71             |                                             |       |                 |       |
| 7      | 57.10 | 7.000            | 57.10                   | 0.875          | 5.250           | 5.125          | 57.29             |                                             |       |                 |       |



# Project- Seamless Pipes & Tubes (Dia 60.3 – 278 mm size )

**Steel Casing Pipe Dimensions and Weight Chart:**

| Lables |       | Outside diameter | Nominal linear mass T&C | Wall thickness | Inside diameter | Drift diameter | Calculated mass C |                                             |       |                 |       |
|--------|-------|------------------|-------------------------|----------------|-----------------|----------------|-------------------|---------------------------------------------|-------|-----------------|-------|
|        |       |                  |                         |                |                 |                | Plain end         | Mass gain or loss due to end finishing (lb) |       |                 |       |
| 1      | 2.00  | D (in)           | lb/ft                   | t (in)         | d (in)          | in             | lb/ft             | Round thread                                |       | Buttress thread |       |
| 1      | 2     | 3                | 4                       | 5              | 6               | 7              | 8                 | 9                                           | 10    | 11              | 12    |
| 7 5/8  | 24.00 | 7.625            | 24.00                   | 0.300          | 7.025           | 6.900          | 23.49             | 15.80                                       |       |                 |       |
| 7 5/8  | 26.40 | 7.625            | 26.40                   | 0.328          | 6.969           | 6.844          | 25.59             | 15.20                                       | 19.00 | 20.60           | 6.21  |
| 7 5/8  | 29.70 | 7.625            | 29.70                   | 0.375          | 6.875           | 6.750          | 29.06             |                                             | 17.40 | 18.80           | 4.41  |
| 7 5/8  | 33.70 | 7.625            | 33.70                   | 0.430          | 6.765           | 6.640          | 33.07             |                                             | 15.80 | 17.00           | 2.61  |
| 7 5/8  | 39.00 | 7.625            | 39.00                   | 0.500          | 6.625           | 6.500          | 38.08             |                                             | 13.60 | 14.60           | 0.21  |
| 7 5/8  | 42.80 | 7.625            | 42.80                   | 0.562          | 6.501           | 6.376          | 42.43             |                                             | 12.01 | 11.39           | -3.01 |
| 7 5/8  | 45.30 | 7.625            | 45.30                   | 0.595          | 6.435           | 6.310          | 44.71             |                                             | 11.04 | 11.04           | -3.36 |
| 7 5/8  | 47.10 | 7.625            | 47.10                   | 0.625          | 6.375           | 6.250          | 46.77             |                                             | 10.16 | 9.23            | -5.17 |
| 7 5/8  | 51.20 | 7.625            | 51.20                   | 0.687          | 6.251           | 6.126          | 50.95             |                                             |       |                 |       |
| 7 5/8  | 55.30 | 7.625            | 55.30                   | 0.750          | 6.125           | 6.000          | 55.12             |                                             |       |                 |       |
| 7 3/4  | 46.10 | 7.750            | 46.10                   | 0.595          | 6.560           | 6.500          | 45.51             |                                             |       |                 |       |
| 7 3/4  | 46.10 | 7.750            | 46.10                   | 0.595          | 6.560           | 6.435          | 45.51             |                                             |       |                 |       |
| 8 5/8  | 24.00 | 8.625            | 24.00                   | 0.264          | 8.097           | 7.972          | 23.60             | 23.60                                       |       |                 |       |
| 8 5/8  | 28.00 | 8.625            | 28.00                   | 0.304          | 8.017           | 7.892          | 27.04             | 22.20                                       |       |                 |       |
| 8 5/8  | 32.00 | 8.625            | 32.00                   | 0.352          | 7.921           | 7.875          | 31.13             | 20.80                                       | 27.60 | 28.30           | 6.03  |
| 8 5/8  | 32.00 | 8.625            | 32.00                   | 0.352          | 7.921           | 7.796          | 31.13             | 20.80                                       | 27.60 | 28.20           | 6.03  |
| 8 5/8  | 36.00 | 8.625            | 36.00                   | 0.400          | 7.825           | 7.700          | 35.17             | 19.40                                       | 25.60 | 26.20           | 4.03  |
| 8 5/8  | 40.00 | 8.625            | 40.00                   | 0.450          | 7.725           | 7.625          | 39.33             |                                             | 23.80 | 24.20           | 2.03  |
| 8 5/8  | 40.00 | 8.625            | 40.00                   | 0.450          | 7.725           | 7.600          | 39.33             |                                             | 23.80 | 24.20           | 2.03  |
| 8 5/8  | 44.00 | 8.625            | 44.00                   | 0.500          | 7.625           | 7.500          | 43.43             |                                             | 21.80 | 22.20           | 0.03  |
| 8 5/8  | 49.00 | 8.625            | 49.00                   | 0.557          | 7.511           | 7.286          | 48.04             |                                             | 19.60 | 19.80           | -2.37 |
| 9 5/8  | 32.30 | 9.625            | 32.30                   | 0.312          | 9.001           | 8.845          | 31.06             | 24.40                                       |       |                 |       |
| 9 5/8  | 36.00 | 9.625            | 36.00                   | 0.352          | 8.921           | 8.876          | 34.89             | 23.00                                       | 32.00 | 31.00           | 6.48  |
| 9 5/8  | 40.00 | 9.625            | 40.00                   | 0.395          | 8.835           | 8.750          | 38.97             | 21.40                                       | 30.00 | 29.00           | 4.48  |
| 9 5/8  | 40.00 | 9.625            | 40.00                   | 0.395          | 8.835           | 8.679          | 38.97             | 21.40                                       | 30.00 | 29.00           | 4.48  |
| 9 5/8  | 43.50 | 9.625            | 43.50                   | 0.435          | 8.755           | 8.599          | 42.73             |                                             | 28.20 | 27.20           | 2.68  |
| 9 5/8  | 47.00 | 9.625            | 47.00                   | 0.472          | 8.681           | 8.525          | 46.18             |                                             | 26.60 | 25.60           | 1.08  |
| 9 5/8  | 53.50 | 9.625            | 53.50                   | 0.545          | 8.535           | 8.500          | 52.90             |                                             | 23.40 | 22.40           | -2.12 |
| 9 5/8  | 53.50 | 9.625            | 53.50                   | 0.545          | 8.535           | 8.379          | 52.90             |                                             | 23.40 | 22.40           | -2.12 |
| 9 5/8  | 58.40 | 9.625            | 58.40                   | 0.595          | 8.435           | 8.375          | 57.44             |                                             | 21.50 | 20.13           | -4.40 |
| 9 5/8  | 58.40 | 9.625            | 58.40                   | 0.595          | 8.435           | 8.279          | 57.44             |                                             | 21.50 | 20.13           | -4.40 |
| 9 5/8  | 59.40 | 9.625            | 59.40                   | 0.609          | 8.407           | 8.251          | 58.70             |                                             |       |                 |       |
| 9 5/8  | 64.90 | 9.625            | 64.90                   | 0.672          | 8.281           | 8.125          | 64.32             |                                             |       |                 |       |
| 9 5/8  | 70.30 | 9.625            | 70.30                   | 0.734          | 8.157           | 8.001          | 69.76             |                                             |       |                 |       |
| 9 5/8  | 75.60 | 9.625            | 75.60                   | 0.797          | 8.031           | 7.875          | 75.21             |                                             |       |                 |       |
| 10 3/4 | 32.75 | 10.750           | 32.75                   | 0.279          | 10.192          | 10.036         | 31.23             | 29.00                                       |       |                 |       |
| 10 3/4 | 40.50 | 10.750           | 40.50                   | 0.350          | 10.050          | 9.894          | 38.91             | 26.40                                       |       | 34.40           | 7.21  |
| 10 3/4 | 45.50 | 10.750           | 45.50                   | 0.400          | 9.950           | 9.875          | 44.26             | 24.40                                       |       | 31.80           | 4.61  |
| 10 3/4 | 45.50 | 10.750           | 45.50                   | 0.400          | 9.950           | 9.794          | 44.26             | 24.40                                       |       | 31.80           | 4.61  |
| 10 3/4 | 51.00 | 10.750           | 51.00                   | 0.450          | 9.850           | 9.694          | 49.55             | 22.60                                       |       | 29.40           | 2.21  |
| 10 3/4 | 55.50 | 10.750           | 55.50                   | 0.495          | 9.760           | 9.625          | 54.26             | 20.80                                       |       | 27.00           | -0.19 |
| 10 3/4 | 55.50 | 10.750           | 55.50                   | 0.495          | 9.760           | 9.604          | 54.26             | 20.80                                       |       | 27.00           | -0.19 |
| 10 3/4 | 60.70 | 10.750           | 60.70                   | 0.545          | 9.660           | 9.504          | 59.45             | 18.80                                       |       | 24.40           |       |
| 10 3/4 | 65.70 | 10.750           | 65.70                   | 0.595          | 9.560           | 9.404          | 64.59             | 16.80                                       |       | 22.00           |       |
| 10 3/4 | 73.20 | 10.750           | 73.20                   | 0.672          | 9.406           | 9.250          | 72.40             |                                             |       |                 |       |
| 10 3/4 | 79.20 | 10.750           | 79.20                   | 0.734          | 9.282           | 9.126          | 78.59             |                                             |       |                 |       |
| 10 3/4 | 85.30 | 10.750           | 85.30                   | 0.797          | 9.156           | 9.000          | 84.80             |                                             |       |                 |       |

API 5 DP Dimensions and Weight Chart:

*Detail Specification for API 5D Drill Pipe*

| Size  | Weight Designation | Calculated Plain-End Weight |       | Outside Diameter |       | Grade   | Wall Thickness |       | Upset Ends. For Weld-on Tools Joint |
|-------|--------------------|-----------------------------|-------|------------------|-------|---------|----------------|-------|-------------------------------------|
|       |                    | Lb/ft                       | kg/m  | in               | mm    |         | In             | mm    |                                     |
| 2 3/8 | 6.65               | 6.26                        | 9.32  | 2.375            | 60.3  | E,X,G,S | 0.280          | 7.11  | EU                                  |
| 2 7/8 | 10.40              | 9.72                        | 14.48 | 2.875            | 73.0  | E,X,G,S | 0.362          | 9.19  | IU or EU                            |
| 3 1/2 | 9.50               | 8.81                        | 13.12 | 3.500            | 88.9  | E       | 0.254          | 6.45  | IU or EU                            |
| 3 1/2 | 13.30              | 12.31                       | 18.34 | 3.500            | 88.9  | E,X,G,S | 0.368          | 9.35  | IU or EU                            |
| 3 1/2 | 15.50              | 14.63                       | 21.79 | 3.500            | 88.9  | E       | 0.449          | 11.40 | IU or EU                            |
| 3 1/2 | 15.50              | 14.63                       | 21.79 | 3.500            | 88.9  | X,G,S   | 0.449          | 11.40 | EU or IEU                           |
| 4     | 14.00              | 12.93                       | 19.26 | 4.000            | 101.6 | E,X,G,S | 0.330          | 8.38  | IU or EU                            |
| 4 1/2 | 13.75              | 12.24                       | 18.23 | 4.500            | 114.3 | E       | 0.271          | 6.88  | IU or EU                            |
| 4 1/2 | 16.60              | 14.98                       | 22.31 | 4.500            | 114.3 | E,X,G,S | 0.337          | 8.56  | EU or IEU                           |
| 4 1/2 | 20.00              | 18.69                       | 27.84 | 4.500            | 114.3 | E,X,G,S | 0.430          | 10.92 | EU or IEU                           |
| 5     | 16.25              | 14.87                       | 22.15 | 5.000            | 127.0 | X,G,S   | 0.296          | 7.52  | IU                                  |
| 5     | 19.50              | 17.93                       | 26.71 | 5.000            | 127.0 | E       | 0.362          | 9.19  | IEU                                 |
| 5     | 19.50              | 17.93                       | 26.71 | 5.000            | 127.0 | X,G,S   | 0.362          | 9.19  | EU or IEU                           |
| 5     | 25.60              | 24.03                       | 35.79 | 5.000            | 127.0 | E       | 0.500          | 12.70 | IEU                                 |
| 5     | 25.60              | 24.03                       | 35.79 | 5.000            | 127.0 | X,G,S   | 0.500          | 12.70 | EU or IEU                           |
| 5 1/2 | 21.90              | 19.81                       | 29.51 | 5.500            | 139.7 | E,X,G,S | 0.361          | 9.17  | IEU                                 |
| 5 1/2 | 24.70              | 22.54                       | 33.57 | 5.500            | 139.7 | E,X,G,S | 0.415          | 10.54 | IEU                                 |
| 6 5/8 | 25.20              | 22.19                       | 33.05 | 6.625            | 168.3 | E,X,G,S | 0.330          | 8.38  | IEU                                 |
| 6 5/8 | 27.70              | 24.21                       | 36.06 | 6.625            | 168.3 | E,X,G,S | 0.362          | 9.19  | IEU                                 |

## End uses of Seamless pipes & tubes

### Applications & Specifications:

#### OIL & GAS SECTOR



API 5L  
API 5CT  
API 5DP

#### HYDROCARBON PROCESS INDUSTRY



ASTM : A-53, A-333, A-334,  
A-335  
BS : 3602 (Pt-I), 3603  
IS : 6286

#### AUTOMOTIVE INDUSTRY



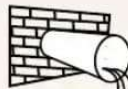
ASTM : A-519  
SAE : 1010, 1012,  
1020, 1040,  
1518, 4130  
DIN : 2391, 1629  
IS : 3601, 3074

#### BOILER, HEAT EXCHANGER, SUPER HEATER & CONDENSER



ASTM : A-106, A-178, A-179,  
A-192, A-209, A-210, A-213,  
A-214  
BS : 3059 (Pt-I & Pt-II),  
EN 10216 (Pt-I & Pt-II)  
3602 (Pt-I)  
IS : 11714  
DIN : 17175

#### WATER & SEWAGE



IS : 1239 (Pt-II), 3589, IS 4270  
BS : 1387  
DIN : 2440, 2441

#### MECHANICAL, STRUCTURAL, & GENERAL ENGINEERING



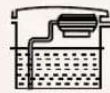
ASTM : A-500, A-501,  
A-519  
DIN : 1629, 1630,  
2391, 17100,  
17200  
BS : EN 10210-1  
EN 10210-2  
EN 10297  
IS : 1161, 3601

#### RAILWAYS



IS : 1239  
(Pt-I), 1161  
RDSO : ETI / OHE / 11

#### WATER WELL



IS : 4270  
ASTM : A-589

#### HYDRAULIC CYLINDER



SAE : 1026, 1518  
IS : 6631  
DIN : 1629

#### BEARING INDUSTRY



SAE : 52100  
DIN : 100 Cr6



### Applications

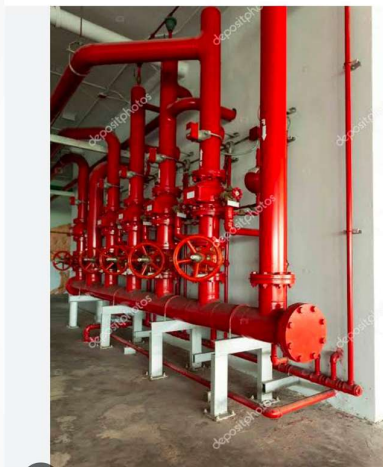


#### Heat exchanger

A heat exchanger is a device or a piece of equipment shaped to transfer any fluid, semi-fluid or gas from one medium to another. The pipes help prevent heat loss while transferring the liquid or gas.

#### Engineering and Construction Industry

Steel seamless pipes are highly used in engineering and construction industries in the following sectors: automobile designing technology, food manufacturing industry, measurement and control technology, hydraulic press, cylinders and pumps, etc. Stainless Steel Seamless pipes are used as a half-finished product in the departments mentioned above. They are used as fittings and tubes installed in big machines or as a half-finished product for machined parts.



#### Chemical and Petrochemical Industries

Steel seamless plants prove highly useful in chemical and petrochemical industries. The area of application in the chemical and petrochemical sectors mainly includes the following: High-pressure pipes in Melamine plants, elaborate piping systems and heat exchanging systems of urea synthesis ethylene plants.

#### Automotive:

- **Exhaust systems:** Their ability to withstand high temperatures makes them ideal for exhaust systems.
- **Suspension components:** They are used in suspension systems due to their strength and durability.
- **Chassis components:** MS seamless pipes are used in various chassis parts.
- **Hydraulic cylinders**



## Applications



### Oil and Gas:

- **Pipeline transportation:** They are used to transport crude oil, natural gas, and other fluids in pipelines, both onshore and offshore.
- **Drilling applications:** They are used in oil and gas drilling operations.
- **Refineries and petrochemical plants:** They are used in various processes within these facilities.

### Power Generation:

- **Boiler tubes:** They are used in boilers to transport high-pressure steam.
- **Condenser and superheater tubes:** They are used in condensers and superheaters in power plants.
- **Turbine and generator components:** They are used in the construction of turbines and generators.

THANK YOU

*-Rank consultant Services*